04 — Análise de Palavras-chave e Mapa Temático

Bibliometria de Gêmeos Digitais para Cidades Inteligentes

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1 Introdução

Este é o notebook mais importante do projeto. A análise de palavras-chave e, especialmente, o **mapa temático** são as peças centrais do artigo, pois revelam:

- Quais temas são dominantes no campo de gêmeos digitais para cidades inteligentes
- Como esses temas se relacionam entre si
- **Onde "decision support" e "public management" estão posicionados** (ou ausentes)

Essa análise é fundamental para sustentar o argumento do gap teórico identificado na pergunta de pesquisa.

2 Carregar Pacotes e Dados

```
library(bibliometrix)
library(tidyverse)
library(igraph)
library(ggraph)
```

```r
library(here)

# Carregar dados processados
df <- readRDS(here("data", "processed", "df_bibliometria.rds"))
cat("Dataset carregado:", nrow(df), "artigos\n")
```

```
Dataset carregado: 671 artigos
```

3 Frequência de Palavras-chave

Antes de construir as redes, vamos examinar quais são as palavras-chave mais frequentes no corpus. Isso dá uma visão geral dos temas presentes.

```r
# Keywords do autor (campo DE) - escolhidas pelos próprios pesquisadores
# Estas são mais relevantes que as Keywords Plus (indexadas automaticamente)

# Extrair e contar keywords do autor
if ("DE" %in% names(df)) {
  kw_autor <- df$DE %>%
    na.omit() %>%
    strsplit(";") %>%
    unlist() %>%
    trimws() %>%
    tolower() %>%
    table() %>%
    sort(decreasing = TRUE) %>%
    as.data.frame()

  names(kw_autor) <- c("Keyword", "Freq")

  # Top 20 keywords do autor
  knitr::kable(
    head(kw_autor, 20),
    caption = "Top 20 palavras-chave do autor (Author Keywords)"
  )
}
```

Table 1: Top 20 palavras-chave do autor (Author Keywords)

Keyword	Freq
digital twin	233
digital twins	109
smart city	72
smart cities	57
artificial intelligence	31
deep learning	27
internet of things	26

Keyword	Freq
urban planning	23
machine learning	22
virtual reality	18
simulation	17
blockchain	16
sustainability	16
digital twin (dt)	15
iot	15
metaverse	15
urban digital twin	14
bim	13
big data	12
internet of things (iot)	12

```r
# Gráfico de barras das keywords mais frequentes
p_kw <- ggplot(head(kw_autor, 20), aes(x = reorder(Keyword, Freq), y = Freq)) +
  geom_col(fill = "#E74C3C", width = 0.7) +
  geom_text(aes(label = Freq), hjust = -0.2, size = 3.5) +
  coord_flip() +
  labs(
    title = "Top 20 Palavras-chave do Autor",
    subtitle = "Digital Twins × Smart Cities (Scopus, 2015-2024)",
    x = NULL,
    y = "Frequência"
  ) +
  theme_minimal(base_size = 13) +
  theme(plot.title = element_text(face = "bold"))

print(p_kw)
```

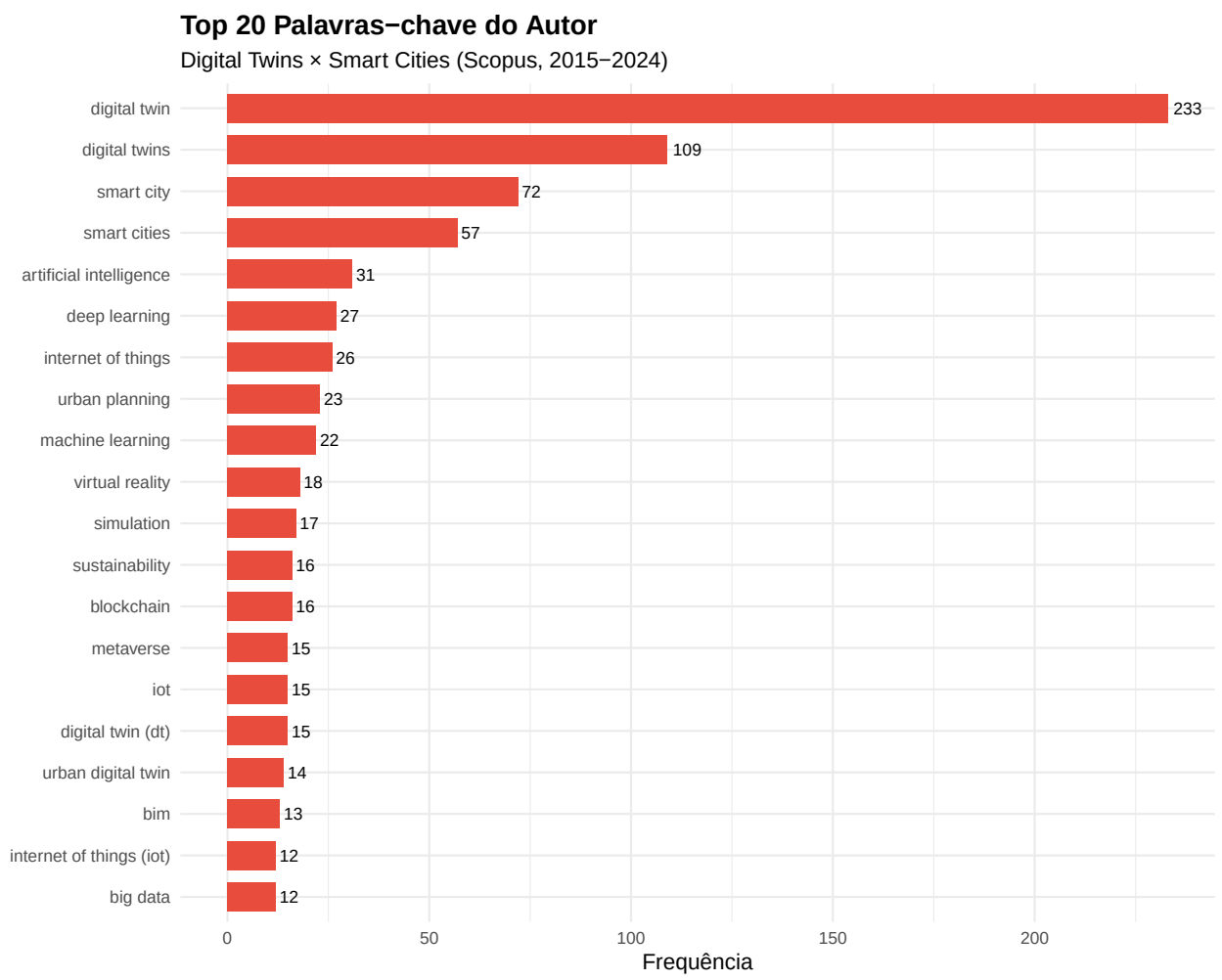

Figure 1: Top 20 palavras-chave do autor mais frequentes

```
ggsave(
  here("output", "figures", "08_top_keywords.png"),
  plot = p_kw,
  width = 10, height = 8, dpi = 300
)
```

```
# ANÁLISE CENTRAL: Verificar presença de termos relacionados ao gap
# Procurar por termos ligados a suporte à decisão e gestão pública

termos_gap <- c(
  "decision support", "decision making", "decision-making",
  "public management", "public administration", "urban governance",
  "governance", "policy", "public policy", "urban management",
  "municipal", "city management", "urban planning decision"
)

cat("=== VERIFICAÇÃO DO GAP: Termos de suporte à decisão e gestão pública ===\n\n")
```

```
=== VERIFICAÇÃO DO GAP: Termos de suporte à decisão e gestão pública ===
```

```r
for (termo in termos_gap) {
  freq <- sum(kw_autor$Keyword == termo)
  if (freq > 0) {
    cat(sprintf("  '%s': %d ocorrências\n", termo, kw_autor$Freq[kw_autor$Keyword == termo]))
  } else {
    cat(sprintf("  '%s': AUSENTE\n", termo))
  }
}
```

```
  'decision support': 5 ocorrências
  'decision making': AUSENTE
  'decision-making': 1 ocorrências
  'public management': AUSENTE
  'public administration': AUSENTE
  'urban governance': 4 ocorrências
  'governance': 2 ocorrências
  'policy': 2 ocorrências
  'public policy': AUSENTE
  'urban management': 5 ocorrências
  'municipal': AUSENTE
  'city management': 1 ocorrências
  'urban planning decision': AUSENTE
```

```r
cat("\n--- Esta análise sustenta (ou refuta) o gap identificado na pesquisa ---\n")
```

```
--- Esta análise sustenta (ou refuta) o gap identificado na pesquisa ---
```

4 Rede de Co-ocorrência de Palavras-chave

A rede de co-ocorrência conecta palavras-chave que aparecem juntas nos mesmos artigos. Clusters nessa rede representam subcampos temáticos.

```r
# Construir a matriz de co-ocorrência de keywords do autor
rede_kw <- biblioNetwork(df, analysis = "co-occurrences", network = "keywords")

cat("Dimensões da matriz de co-ocorrência:", dim(rede_kw), "\n")
```

```
Dimensões da matriz de co-ocorrência: 4121 4121
```

```r
# Visualizar a rede de co-ocorrência
set.seed(2024)

net_kw <- networkPlot(
  rede_kw,
  n = 40,                     # Top 40 keywords
  type = "fruchterman",
  Title = "Co-ocorrência de Palavras-chave do Autor",
```

```
  size.cex = TRUE,
  label.cex = TRUE,
  edgesize = 3,
  edges.min = 3,
  community.repulsion = 0.1
)
```

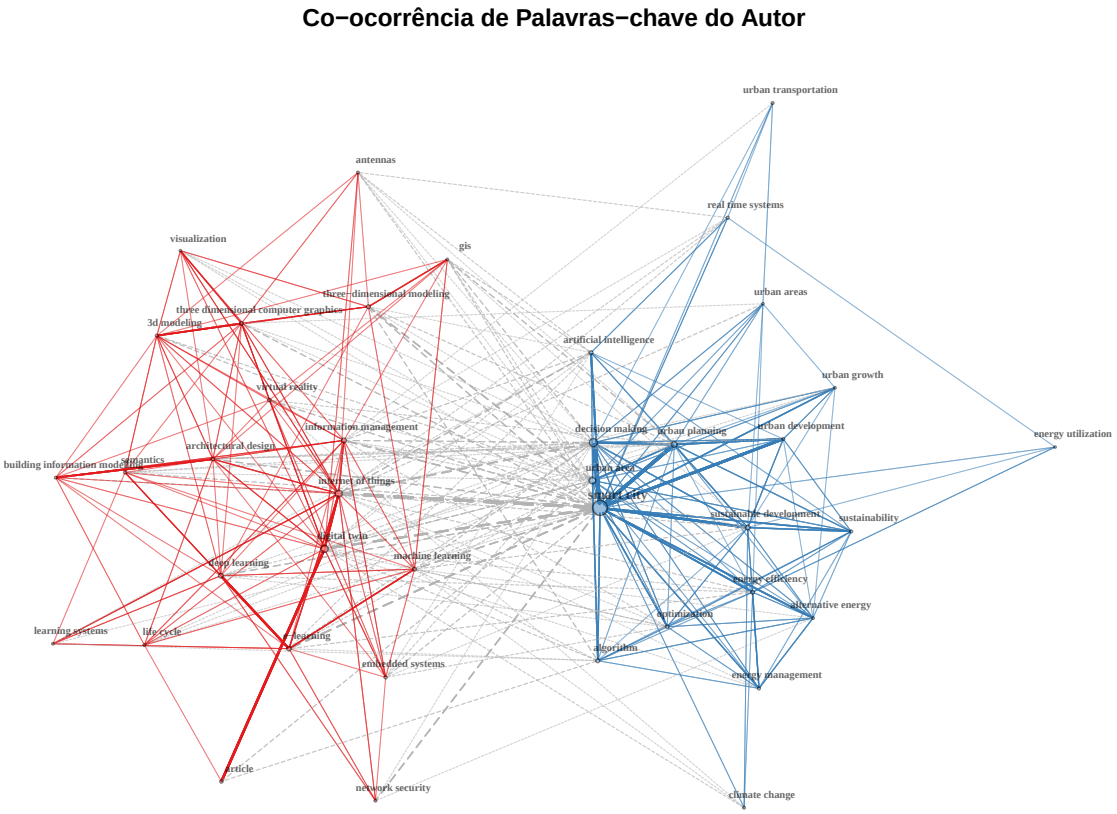

Figure 2: Rede de co-ocorrência de palavras-chave do autor

```
# Salvar a rede
png(here("output", "figures", "09_coocorrencia_keywords.png"),
    width = 10, height = 8, units = "in", res = 300)

set.seed(2024)
networkPlot(
  rede_kw,
  n = 40,
  type = "fruchterman",
  Title = "Co-ocorrência de Palavras-chave do Autor",
```

```r
  size.cex = TRUE,
  label.cex = TRUE,
  edgesize = 3,
  edges.min = 3,
  community.repulsion = 0.1
)
```

```
$graph
IGRAPH fe6fa75 UNW- 40 1665 --
+ attr: alpha (g/n), ylim (g/n), xlim (g/n), rescale (g/l), asp (g/n),
| layout (g/n), main (g/c), name (v/c), deg (v/n), size (v/n),
| label.cex (v/n), color (v/c), community (v/n), label.dist (v/n),
| frame.color (v/c), label.color (v/c), label.font (v/n), label (v/c),
| num (e/n), width (e/n), color (e/c), lty (e/n), weight (e/n), curved
| (e/l)
+ edges from fe6fa75 (vertex names):
[1] smart city--decision making smart city--decision making
[3] smart city--decision making smart city--decision making
[5] smart city--decision making smart city--decision making
+ ... omitted several edges

$graph_pajek
IGRAPH 660c211 UNW- 40 2123 --
+ attr: name (v/c), deg (v/n), size (v/n), label.cex (v/n), id (v/c),
| num (e/n), weight (e/n)
+ edges from 660c211 (vertex names):
 [1] smart city--decision making smart city--decision making
 [3] smart city--decision making smart city--decision making
 [5] smart city--decision making smart city--decision making
 [7] smart city--decision making smart city--decision making
 [9] smart city--decision making smart city--decision making
[11] smart city--decision making smart city--decision making
[13] smart city--decision making smart city--decision making
+ ... omitted several edges

$cluster_obj
IGRAPH clustering walktrap, groups: 2, mod: 0.18
+ groups:
  $`1`
   [1] "internet of things"
   [2] "digital twin"
   [3] "deep learning"
   [4] "information management"
   [5] "virtual reality"
   [6] "e-learning"
   [7] "three-dimensional modeling"
   [8] "machine learning"
   [9] "three dimensional computer graphics"
```

```
  + ... omitted several groups/vertices

$cluster_res
                                vertex cluster btw_centrality clos_centrality
4                   internet of things       1   104.59819319     0.020534503
5                         digital twin       1    93.05060261     0.022175683
6                        deep learning       1     6.55087988     0.018690200
10              information management       1    54.45721430     0.020040125
11                     virtual reality       1     6.94925768     0.018279752
12                          e-learning       1    13.71891852     0.019119504
13          three-dimensional modeling       1    17.07485314     0.019119504
17                    machine learning       1    46.64911740     0.020040125
20 three dimensional computer graphics       1     7.05483298     0.018279752
22                architectural design       1    33.16861405     0.017886944
23                    embedded systems       1     4.08282595     0.018279752
26                           semantics       1     2.01367808     0.018279752
27                                 gis       1    10.93250616     0.019119504
28                       visualization       1     0.39694443     0.017886944
29                         3d modeling       1     0.65287272     0.017510663
31                             article       1     0.60485224     0.015252982
32                            antennas       1     6.43447830     0.019119504
33      building information modelling       1     0.00000000     0.016803676
34                    learning systems       1     1.09159293     0.017886944
35                    network security       1    10.86451932     0.017510663
38                          life cycle       1     0.65324614     0.017690017
1                           smart city       2   368.35127907     0.027259170
2                       decision making       2    60.18088321     0.022782636
3                        urban planning       2    12.04160284     0.018265278
7                            urban area       2    59.19053402     0.021873799
8              sustainable development       2     6.92943303     0.018074131
9              artificial intelligence       2    15.72961840     0.018894673
14                   urban development       2     0.11515469     0.013503078
15                   energy efficiency       2     5.13689519     0.016968773
16                           algorithm       2    12.57460046     0.017149886
18                   energy management       2     0.24797240     0.015990823
19                      sustainability       2     0.00000000     0.010779201
21                      climate change       2     0.58834570     0.015844126
24                        optimization       2     4.16759361     0.017886944
25                   real time systems       2     3.12031819     0.016803676
30                        urban growth       2     0.07966916     0.015833250
36                urban transportation       2     0.00000000     0.013860635
37                  alternative energy       2     0.27687010     0.014830630
39                         urban areas       2     0.26922993     0.014978511
40                  energy utilization       2     0.00000000     0.009767328
   pagerank_centrality
4          0.046716015
5          0.040951738
6          0.027795469
```

```
10          0.029307576
11          0.017157592
12          0.028805128
13          0.021576443
17          0.023378653
20          0.029389411
22          0.030121686
23          0.017313053
26          0.017498602
27          0.016147294
28          0.016925785
29          0.027321103
31          0.012251871
32          0.011758818
33          0.020453909
34          0.012985875
35          0.011727477
38          0.014485388
1           0.093730271
2           0.047946239
3           0.040928469
7           0.036798821
8           0.036992398
9           0.021671382
14          0.026354126
15          0.029458026
16          0.023204148
18          0.021365138
19          0.025409728
21          0.009032875
24          0.024715724
25          0.012368158
30          0.017052517
36          0.008840499
37          0.025945509
39          0.015146895
40          0.008970192

$community_obj
IGRAPH clustering walktrap, groups: 2, mod: 0.18
+ groups:
  $`1`
   [1] "internet of things"
   [2] "digital twin"
   [3] "deep learning"
   [4] "information management"
   [5] "virtual reality"
   [6] "e-learning"
```

```
   [7] "three-dimensional modeling"
   [8] "machine learning"
   [9] "three dimensional computer graphics"
  + ... omitted several groups/vertices

$layout
             [,1]        [,2]
 [1,]  0.09164924 -0.14886121
 [2,]  0.07855870  0.03640039
 [3,]  0.23982023  0.03101149
 [4,] -0.42983537 -0.10844578
 [5,] -0.45817631 -0.26535633
 [6,] -0.66490494 -0.34147689
 [7,]  0.07665251 -0.07131870
 [8,]  0.38633729 -0.20481933
 [9,]  0.07375740  0.29167788
[10,] -0.41944016  0.04243613
[11,] -0.56823296  0.15681101
[12,] -0.52897832 -0.54855225
[13,] -0.37053441  0.42193441
[14,]  0.45744516  0.04654156
[15,]  0.39625023 -0.38841823
[16,]  0.08716593 -0.58273338
[17,] -0.27847249 -0.32295766
[18,]  0.40892221 -0.66143378
[19,]  0.59285111 -0.21587541
[20,] -0.62388243  0.37465724
[21,]  0.37924398 -1.00000000
[22,] -0.68095129 -0.01069502
[23,] -0.33612891 -0.63012306
[24,]  0.22576116 -0.48642240
[25,]  0.34654567  0.67450693
[26,] -0.85635276 -0.04843976
[27,] -0.21303633  0.55608003
[28,] -0.74567542  0.58099606
[29,] -0.79277363  0.33968280
[30,]  0.56062158  0.19203060
[31,] -0.66385725 -0.92538545
[32,] -0.39140076  0.80282280
[33,] -0.99449023 -0.06311853
[34,] -1.00000000 -0.53465765
[35,] -0.35640808 -0.98006179
[36,]  0.43614212  1.00000000
[37,]  0.51675956 -0.46185261
[38,] -0.81766030 -0.53825185
[39,]  0.41684981  0.43029650
[40,]  1.00000000  0.02359060
```

```
$S
NULL

$nodeDegree

smart city
decision making
digital twin
internet of things
urban area
urban planning
information management
sustainable development
deep learning
e-learning
algorithm
three-dimensional modeling
artificial intelligence
energy efficiency
three dimensional computer graphics
machine learning
architectural design
virtual reality
article
optimization
embedded systems
real time systems
sustainability
semantics
energy management
3d modeling
urban development
urban transportation
alternative energy
climate change
network security
antennas
building information modelling
urban growth
learning systems
life cycle
gis
urban areas
visualization
energy utilization
building
forecasting
digital storage
```

human
block-chain
data visualization
interoperability
data handling
blockchain
decision support systems
machine-learning
unmanned aerial vehicles (uav)
smart power grids
china
data driven
cyber physical system
computer simulation
information theory
real- time
real-world
smart grid
intelligent systems
risk assessment
interactive computer systems
big data
data acquisition
real - time system
security
geographic information systems
digitization
urbanization
urban environments
numerical model
optical radar
urban design
point-clouds
mapping
case-studies
aerial vehicle
city model
disasters
water management
data set
metaverses
floods
vehicles
large-scales
optimisations
edge computing
built environment
artificial neural network

data integration
urban digital twin
roads and streets
construction industry
intelligent buildings
water supply
object detection
5g mobile communication systems
buildings
land use
controlled study
remote sensing
computational modelling
adult
behavioral research
unmanned aerial vehicle
automation
modeling
cost effectiveness
renewable energies
uncertainty
quantitative analysis
ontology
strategic approach
sensor
intelligent transportation systems
learning algorithms
spatial data
economic and social effects
data analytics
simulation
traffic congestion
stakeholder
geo-spatial
internet
3d-modeling
flood control
traffic management
real time
software
energy
long short-term memory
economics
urban transport
population statistics
benchmarking
'current
monitoring

semantic segmentation
floodings
autonomous vehicles
metadata
urban drainage
stochastic systems
sustainable development goal
3d models
technological development
united states
city
road
urban buildings
renewable energy
humans
proof of concept
emission control
augmented reality
australia
decision support system
carbon dioxide
comparative study
conceptual framework
resource allocation
costs
data mining
physical objects
office buildings
urban drainage systems
risk management
information modeling
ecosystems
experimental study
innovation
water quality
microgrid
drainage
mean square error
real time control
urban management
motor transportation
environmental technology
industry 4.0
digital devices
intelligent vehicle highway systems
network architecture
construction
database

performance
image reconstruction
transportation system
deep neural networks
solar energy
flooding
prediction
sewer
bayesian networks
computer crime
runoff
air quality
intelligent transportation
spatial analysis
local participation
environmental management
hydrodynamics
cost benefit analysis
data fusion
graph neural networks
virtual spaces
power
data processing
lidar
wireless communications
particle swarm optimization (pso)
urban energy
three-dimensional display
engines
cyber-physical systems
storms
energy use
iterative methods
pipelines
optimization algorithms
cyber security
energy management systems
melbourne
victoria [australia]
disaster prevention
reinforcement learning
cybersecurity
retrofitting
internet of thing
street traffic control
multi agent systems
building energy model
cyber physicals

integration
pedestrian
city scale
rain
uncertainty analysis
information use
italy
three dimensional displays
multiobjective optimization
technology adoption
demand-side management
virtual representations
performance assessment
urban systems
model predictive control
simulation platform
knowledge graphs
intrusion detection
ontology's
detection method
millimeter waves
open data
semantic web
hydrological modeling
city digital twin
3d visualization
crime
digital replicas
historic preservation
cybe-physical systems
mapping method
neural networks
immersive
intelligent transportation system
3d reconstruction
solar radiation
energy model
solid modelling
computer vision
iot
3d city model
convolutional neural network
information system
enabling technologies
genetic algorithms
housing
data collection
sensors data

geographic information system
graphic methods
real-time control
traffic signals
spatiotemporal analysis
building information modeling
data privacy
internet of things (iot)
future prospect
physical systems
autonomous driving
autonomous agents
work-flows
infrastructure
cloud-computing
emerging technologies
information and communication technology
mobile communications
landscape design
supply chains
robotics
traffic flow
building model
large datasets
multi-objectives optimization
open source software
computer architecture
decisions makings
energy-consumption
industrial research
carbon emissions
environmental monitoring
pipeline
stochastic models
mobile telecommunication systems
reliability analysis
resource management
roofs
time series analysis
errors
urban drainage model
water supply systems
sewage
5g
carbon emission
electric power distribution
knowledge graph
resources allocation

technological advancement
urban air
calibration
carbon
digital technologies
% reductions
drainage network
crowdsourcing
construction equipment
job analysis
systematic review
task analysis
water
geometry
hydrology
water levels
digital transformation
implementation process
adaptive control systems
building information model
building information model - bim
economic development
efficiency
energy conservation
twin networks
technology
digital twin city
internet of vehicle
denmark
information analysis
vehicle to vehicle communications
covariance matrix
eco routing
fog
systematic literature review
predictive control systems
autonomous vehicle
governance approach
infrastructure systems
segmentation
cities
engineering education
fossil fuels
localisation
sensor networks
electric loads
photovoltaic system
social networking (online)

complex networks
facilities management
model and simulation
decision supports
humidity control
cross-domain
digital elevation model
information systems
spain
reconstruction
high resolution
architecture
drones
solar power generation
urban morphology
in-buildings
rapid urbanizations
knowledge
mixed reality
budget control
quality control
renewable energy integrations
solar power
textures
computer software
manufacturing
electric load dispatching
structural optimization
urban mobility
airflow
agriculture
classification (of information)
disaster management
digitalization
infrastructure planning
recurrent neural networks
stormwaters
zero emission
simulation model
videorecording
decarbonisation
real time monitoring
conceptual model
support vector machines
bayesian analysis
hydraulic models
models
theoretical

theoretical model
wind power
traffic simulations
urban resilience
catchments
convolution
cost reduction
highway planning
image fusion
water distribution systems
nondestructive examination
scheduling
literature reviews
planning
renewable resource
microclimate
open systems
thermography
digital mapping
decentralised
federated learning
fault detection
smart solutions
travel time
water networks
humidity
markov chain
monitoring system
project management
quality of service
greenhouse gases
laser applications
mesh generation
tunnel
urban road
airborne lidar
netherlands
accuracy assessment
light rail transit
pressung
reinforcement learnings
building stocks
large scale systems
photogrammetry
resilience
motion planning
wireless sensor networks
condition assessments

measurement accuracy
health care
city information modeling
integrated approach
mobile mapping systems
waters managements
probability density function
traffic and transport
low-costs
neighbourhood
simple++
south korea
inspection
greenspace
clustering algorithms
electric vehicles
investments
urban building energy modeling
wireless networks
nature-based solution
air conditioning
dynamics
fog computing
geo-spatial data
potential energy
sustainable energy
water budget
maintenance
infrastructural development
urban infrastructure
digital twin technology
electric power transmission networks
inference engines
objects detection
public-private partnership
traffic control
neighborhood
virtual environments
5g mobile communication
authentication
energy systems
environmental impact
structural health monitoring
helsinki
waste management
6g mobile communication
brain
data quality

energy efficient
cyber-attacks
error analysis
pluvials
urban scale
high-fidelity
industrial robots
knowledge management
trajectories
vehicle transmissions
3d point cloud
accidents
city planning
convolutional neural networks
kinematics
literature review
magnetic levitation vehicles
operations and maintenance
software design
barcelona
blockchain technology
management frameworks
nearest neighbor search
sensor nodes
public health
atmospheric pollution
cameras
computer aided design
data centric
emergency medical services
frequency modulation
integer programming
modeling data
security of data
weather
data assimilation
aircraft detection
predictive models
scenario analysis
state of the art
finance
intrusion detection systems
nitrogen oxides
remote-sensing
advanced technology
air mobility
level-of-detail
medical services

object recognition
unmanned vehicle
hong kong
participatory approach
bibliometric
building facades
carbon footprint
digital representations
ecosystem
sustainable city
estonia
sustainable cities
computational fluid dynamics
peer to peer networks
software testing
analytical method
exhaust gas
flood
gluing
markup languages
photovoltaics
public/private partnerships
vibration
water conservation
water pollution
air pollutant
air pollutants
air pollution control
blurred vision
car driving
clinical article
digital twin platform
dizziness
driving strategy
eco-driving
emissions mitigation
emissions monitoring
eye disease
eyestrain
fatigue
female
headache
male
motor vehicle
motor vehicles
nausea
nitrogen oxide
on-road emissions

prevention and control
procedures
routing algorithms
self report
traffic conditions
vehicle emission
vehicle emission mitigation
vehicle emissions
vertigo
artificial intelligence algorithms
clean energy
metropolitan area
safety management
cloud analytics
cultivation
data interoperability
false data injection attacks
household energy
multi criteria decision-making
multicriteria decision-making
multicriterion decision makings
ontology-based
pollution control
product life cycles
renewable energy source
risk perception
road planning
water use
cadastre
diagnosis
distributed computer systems
diverse applications
energy planning
linked data
ant colony optimization
framework
population growth
regulatory compliance
volunteered geographic information
digital image
embeddings
lidar point cloud
user interfaces
3d data fusion
bibliometrics analysis
computational methods
data-driven approach
datum transformation

hydrodynamic modeling
particle swarm
particle swarm optimization
swarm optimization
travel behavior
unit-based
urban energy systems
advanced informations
conceptual frameworks
graph theory
6g
city management
crop growth
cyber physical systems (cpss)
demand response
energy flexibility
energy-integration
health risks
policy making
data driven decision
decision makers
geographic information
river pollution
aerial survey
crops
data management
industry 5.0
kalman filters
kalman-filtering
maintenance operations
network theory (graphs)
rural areas
sensors network
sewer network
stochastics
surface water
surface waters
atmospheric modeling
excavation
solar panels
adaptive boosting
binary trees
emergency services
optimal scheduling
peak load
resource recovery
secondary batteries
solar concentrators

source term estimation
stormwater
urban intelligence
waste water treatment plants
adaptivity
air monitoring
atmospheric dispersion
atmospheric transport
bayesian approaches
bayesian learning
bayesian solution
canopy
complex atmospheric environments
concentration (parameter)
dispersion models
end-users
environmental temperature
error diagnosis
estimation method
fluid intake
flushing
fyn
geographic elevation
hydrological model
importance sampling
measurement error
model validation
monte carlo analysis
monte carlo method
noise measurement
odense
parameter estimation
plume dispersion
pollutant source
pollution sources
rain uncertainty
sequential analysis
sequential importance sampling
sequential monte carlo
soil moisture
source parameters
statistical model
storm water
syddanmark
synthetic example
three-dimensional imaging
time factor
times series

validation
variance
water flow
water level
water level sensors
water movements
wind speed
anomaly detection
information and communication technologies
modeling approach
urban sustainability
adaptive traffic signal control
automatic detection
city construction
computational geometry
coordinate
district heating
earth observations
environment monitoring
fires
hangzhou
internet of things technologies
pavements
sustainable management
urban and rural
urban regeneration
world wide web
zhejiang
quality of life
germany
digital infrastructures
arcgis
attack detection
bayesia n networks
condition
controllers
emergency management
energy hubs
flood assessment
france
ile de france
leakage detection
model visualization
optimizers
paris
piedmont [italy]
study areas
sustainable smart city

sustainable urban development
ville de paris
gnss
image classification
digital hydraulics
finland
healthcare systems
numerical models
pumps
quality management
realtime simulation (rts)
varsinais-suomi
vehicle network
3-d mapping
citygml
cluster analysis
earth (planet)
ferroelectric ram
gases
global positioning system
markov processes
multi-modal
photomapping
physical products
pv system
stereo vision
acoustic inspection
acoustic technology
defects inspections
deterioration
echolocation
ecosystem monitoring
electromagnetic radiation
environmental parameters
fiber optics
magnetic field
microwave radiation
monitoring approach
petrochemical industry
pipe
pipe deterioration
quantitative evaluation
sewage pipes
sewer pipelines
sewers
stereo image processing
stereolithography
telecommunication

ultrasound
visual inspection
waste water management
satellite data
physical modelling
assessment method
cellular network
emergency health service
energy savings
energy-savings
feature extraction
forecasting method
interpolation
mobility vehicles
online systems
satellites
situational awareness
velocity
data structures
historical data
3d city modeling
airborne laser scanning
building energy
coal mines
computer software reusability
digital libraries
distribution system
electronic data interchange
energy harvesting
energy resource
extraction
fault tolerance
first responders
genetic algorithm
geo-spatial informations
health risk
integrated management
intelligent robots
light detection and ranging
light emitting diodes
mobile phone
prediction modelling
sustainable development goals
vehicle's dynamics
whale optimization algorithm
aggregation
austria
evaporation

fluid balance
fully automated
graz
infiltration
innovative method
land cover
pruning
skeletonization
sparsification
storm water management model
streamline
styria
surface runoff
urban population
human-in-the-loop
urban policy
accident prevention
automobile testing
city logistics
computer graphics
data communication systems
electric potential
global warming
land use change
mm waves
multi agent
near-real time
sea level
sustainable urban systems
perception
aerial photography
agent based simulation
agricultural robots
behavioral science
city information model
computer programming
construction projects
deep reinforcement learning
demolition
disruptive technology
internet of thing (iot)
maturity levels
mobile computing
mobile-computing
model-based opc
pollution exposure
regional planning
statistics

taxonomies
technological solution
topography
urban air mobility
video streaming
water planning
boston
building characteristics
building semantic
computer aided instruction
construction period
data availability
feasibility study
massachusetts
morphology
satellite imagery
environmental protection
factor analysis
greenhouse gas
greenhouses
highway administration
industrial internet of thing
industrial sector
location
measurements of
peer-to-peer computing
physical environments
transportation infrastructures
zero energy buildings
virtual worlds
agent based
artificial intelligence technologies
current data
data messages
dataflow
distributed ledger
ecology
energy policy
information communication technology
logistic operations
open-source
system of systems
urban transportation networks
birds
cesium ion
city geometry markup language
database management system
industry foundation class

information model
markup language
materials management
resource description framework (rdf)
resources description frameworks
transformation methods
web services
thailand
urban forestry
bridges
building energy performance
coastal zone
data preparation
data transfer
digital twins
freight transportation
infrastructure managements
knowledge based systems
landscape structure
leak detection
leakage localization
level fusion
modeling analyzes
natural gas
participatory sensing
particulate matter
predictive analytics
regression analysis
scalability
smart campus
smart contract
water treatment
control strategy
feedback system
hybrid measure
irrigation
irrigation (agriculture)
irrigation and drainage systems
irrigation system
irrigation systems
meteorology
natural science
real time control systems
retention areas
water depth
ad-hoc networks
application research
collaboration

computer-aided design
edge intelligence
georeferencing
highway traffic control
last mile
oblique images
prioritization
rock mechanics
traffic emission
urban agriculture
urban system
varanidae
vehicular networks
decision trees
development and applications
energy internet
green buildings
health care education
management systems
object tracking
profitability
quality-of-service
remote control
systems analysis
united kingdom
university campus
urban problems
virtual addresses
web3
computation offloading
mobile edge computing
binary alloys
collection systems
college buildings
data platform
drone
holograms
human computer interaction
information services
license plates (automobile)
management platforms
model buildings
new forms
operating costs
potassium alloys
proposal
reference architecture
system-of-systems

uranium alloys
what-if analysis
cloud computing
adversarial machine learning
ageing infrastructures
air navigation
current research status
facades
holistic approach
low emission
mobile mapping
synchronization
task offloading
traffic loads
vehicular edge computing
walkability
wastewater treatment
circular bioeconomy
data pipelines
emerging contaminant
intelligent data
operations and control
process design
sewage pumping plants
wastewater reclamation
wastewater treatment plant
water availability
water recycling
water treatment plants
economic analysis
image analysis
valencia
business logic
business-intelligence
electric bicycles
electric bikes
electricity generation
energy transitions
industrial context
kalman filter
model algorithms
policy implementation
prototype system
research gaps
spatial informations
time synchronization
traffic signs
urban politics

control systems
cybe-physical digital twin
cybersecurity enhancement
deep learning framework
digital control systems
dynamic mapping
grid control
information processing
land use type
learning frameworks
nearest neighbor analysis
philosophy
resource constraint
resource development
rural ecological landscape
rural landscape
smart grid control system
space constraints
support vector machine
food production
low latency
management is
mitigation
mobility
physical assets
sampling
scientometric analysis
tree
urban community
urban spaces
3d inundation mapping
assessment of environmental satisfaction
beijing [beijing (ads)]
beijing [china]
biological systems
building digital twin
building information model big data
citizen's participations
city designs
controlled environment
controlled environment agricultures
crop production
crop yield
crowdsourcing platforms
cutting edge technology
data fusion algorithm
data value extension
data values

dynamic host configuration protocols
efficiency measurement
efficient managements
energy efficient building
energy optimization
environmental assessment
environmental factor
environmental protection agency
evolutionary optimization techniques
evolutionary optimizations
flood modeling
hydrological cycle
integrated optimization
inundation mappings
matlab
model-predictive control
models integration
physic-informed deep learning
plant factory
protocol cans
railway transport
rapid flood spreading model
resource sustainability
smart urban energy system
supervision
sustainable city design
traditional knowledge
urban rail transit
virtual reality simulations
water resource
application scenario
data informations
decision analysis
human-centric
real-time information
surveys
agent-based model
bandwidth
chaoyang
coverage extension
dynamic disaster risk
economic growths
frequency analysis
image communication systems
intelligent reflective surface
liaoning
nanning
pedestrian flow

pedestrian mobility
reflective surfaces
risk levels
sponge city
unmanned aerial vehicle cellular network
waterlogging
3d maps
3d representations
attica
benchmark
citizen science
coronavirus
covid-19
distribution grid
greece
international standards
local government
mobility systems
seamless integration
technology-based
twin techniques
urban sensing
car use
climate adaptation
complex production
convolutional generative adversarial network
design
economic progress
energy prices
environmental risk
europe
fleet operations
flood risks
generative adversarial networks
green bond
imagery
jiangsu
low consistency
nanjing [jiangsu]
optimization procedures
oriented systems
phase 1
pluvial flood-ing
process consistency
production process
routing decisions
routing services
shanghai

six-phase
smart decision making
solar cells
spatial planning
stormwater-resilient urban design
sweden
time cost
urban energy transaction
vehicle fleets
air traffic control
construction method
data science
digital integrated circuits
distributed simulations
energy simulation
low income population
sensing devices
topology
4d spatial data management
aiot platform
alignment selection
analog storage
ann
apriltag
autonomous navigation
building demolition
bus sensitivity ratio
cyber attacks
data mining for urban building energy modeling
detection models
digital twin of historic cluster
directed graphs
dragonfly optimization technique
driving route selection habit
electric power transmission
electric towers
energy assessment
energy networks
energy retrofit
four-dimensional modeling
gbdttag
generator sensitivities
generator sensitivity ratio
graphic-based digital twin
heritage building information modeling
heritage buildings
historical record
hybrid techniques

image acquisition
image coding
image matching
image texture
imc
infrared imagery
keras
lca
localization and pose estimation
microcontrollers
multi-scales
network layers
neuromorphic computing
noise
observational method
online bim
openthread
optimization problems
optimization techniques
pilot applications
pose-estimation
renewable energy resources
reproducibility
reproducibility of results
route selection
science based decision makings
snn
spatial data management
statistical analysis
statistical tests
structural response
sustainable urban road planning
tensorflow
tensorflow for microcontroller
tensorflow lite
tinyml
torino [piedmont]
turin
vehicle localization
water towers
zephyr
building blockes
communications systems
critical issues
manufacture
potential benefits
sociotechnical
sustainable building

3-d digital modeling
adaptive maintenance
adaptive systems
adjoint problems
adjoints
apache spark
automated detection
automated tracking
backend system
barcelona [barcelona (prv)]
barcelona [catalonia]
berkelium
building and heating permit
caching
canada
carbon monoxide
catalonia
civil defense
cloud-in-the-loop simulation
coastal urban flood risk management
cognitive workloads
communication hub
comprehensive monitoring
computer software portability
computer system
computing and communication hub
construction information
control capabilities
cortisol
cultural diversity
culture
data offloading
data transaction
decentralized finance
delphi analysis
delphi technique
depolluting panel
detection and tracking
digital innovations
digital twin network
disaster
einsteinium
electric discharges
electric vehicle
electric vehicle (electric vehicle)
emergency management cycle
emergency responders
energy mapping

energy system digital twin
facilities management process
flood risk management
fundings
garages (parking)
gas emissions
geometric deep learning
green funding
heating
integration testing
interactive planning
kubernetes
lawrencium
learning methods
machine learning and reasoning
management cycle
management economics
meta model
microgrid architecture
multi micro-grids
municipal housing retrofit program planning
nginx
numerical strategies
oil tanks
operational digital twin
orbital laboratories
orbital transfer
outdoor air qualities
participatory planning
participatory planning and governance
personal computing
persons with disabilities
photoelectrochemical cells
photovoltaic modules
plutonium
power quality
program planning
real case
real-case study
realistic model
rescue personnel
retrofit programs
sailing vessels
sea level change
sensor configurations
sensor placement
simulation input
smart thermostat data

social-economic
software aging
software rejuvenation
spanish urban agenda
stochastic secure framework
stochasticity
sulfur dioxide
systematic workflow
thermal
thz
traffic pollution
unmanned surface vehicles
urban digitalization
urban facility management
urban flood risks
virtual storage
zoning
achilles tendon
cell-free
comunidad valencia
democracy
musculoskeletal system
neurorehabilitation
access to information
aerial oblique imagery
aerial photograph
agriculture engineering
airborne laser scanning point cloud
airborne survey
automatic building reconstruction
automatic buildings
autonomous software agents
backpack
biomedical monitoring
brussels capital region
building reconstruction
chimneys
clinical application
communication capabilities
computational capability
computational fluid
computational fluid dynamic
computer model
connected corridor
data silos
deep integrations
digital twin based healthcare system
distributed power generation

electric energy storage
energy storage
environmental performance
flood risk assessments
flow visualisation
fluid-dynamics
geometry preparation
global trends
ground based
ground-based measurement
holistic perspectives
identity and access management
identity and access managements
industrial standards
informatics standard
international organization for standardization
international standards organization
islanding
iso standards
k-nearest neighbors classifiers
laser
laser scanning point clouds
life-cycle information
medical applications
medical diagnostic imaging
medical imaging
memory architecture
mhealth
microgrid systems
mobile mapping system
model library
motion compensation
nearest neighbours method
neurological rehabilitation
optimal fusion
parkinson disease identification
parkinson's disease
partitioning algorithms
partitions (building)
pattern recognition
performance measure
primitive model
primitive model library
product lifecycle information management
real-world information
reconstruction (structural)
reconstruction optimization
remote patient monitoring

security architecture
shape optimization
smart city and virtual care application
smartphones
solar and wind renewable energy resource
solar and winds
standard
storage systems
structure (composition)
structure-aware
structure-aware partitioning algorithm
sustainable urban management
tendons
tie points
time of use
traffic flow visualization
traffic monitoring
traffic performance
traffic sensor network
traffic sensors
united nations
urban wind flows
virtual assistants
academic research
dataset
digital modeling
geotechnical engineering
smart infrastructures
traffic demands
3d data
agricultural robotics
algae control
all wheel drive vehicles
automobile suspensions
basilicata
bayesian
bayesian machine learning
bayesian network structural learning algorithm
buried pipes
co2 emissions
colombia
combinatory inference
complex data
computer graphics equipment
data-centric geotechnic
data-driven site characterization
data-informed decision support index
digital hydraulic model

dissimilarity matrix
electromechanical equipments
energy digitalization
energy needs
energy resilience
equipment
equipment maintenance
features recognition
fertilizers
field robotics
fitting primitive
four wheel steering
geotechnics
graphics processing unit
green infrastructure
heading directions
hierarchical clustering
hierarchical clustering analysis
human resource management
hydraulic analysis
image processing
informed decision
internet of energy
junction
knowledge ontology
leakage
lithium compounds
load management
matera
mobility over deformable terrain
model automobiles
multi-body modelling
multibody simulations
multimedium big data
multimodal transformer
multisensory
navarra
off road vehicles
off-road vehicles
operation status
pamplona
power management
preprocessing tools
primitives recognition
proactive maintenance
program processors
project deepgeo
reasoners

recognition
reinforcement learning approach
remote sensing technology
research topics
road vehicles
robot applications
robot programming
robot simulation framework
robot simulations
rule extraction and updating
rules extraction
safety
safety data
safety management systems
semantic web technology
sensors data fusion
simulation framework
site characterization
smart construction
smart water management
soil property
sports
stratigraphy
structural learning
support vector machine (svms)
tallinn
temperature control
unstructured point clouds
unsupervised method
urban excavation safety
vehicle wheels
virus disease
waste heat
water distribution networks
water usage
water usage prediction
biomechanics
historic building
ieee standards
landscape planning
smoke
advanced analytics
advection-reaction-diffusion equations
application programming interfaces (api)
artificial bees
aspect ratio
austin [texas]
basin management

bearing capacity
big earth data
branch union
branches
breaking stress
building energy flexibility
capacity
city river
cloud-fog computing
cnn-lstm
combined-heat and power
computational framework
constant stress
contaminant transport modeling
contaminants transport
contamination
continuous modeling
control system
cost engineering
covariance matrix adaptation evolution strategies
critical component
data-model fusion dispatch
deep impact
demand analysis
design method
differentiation
digital ecosystem
digital twin modeling
dispatch strategy
drainage networks
economic dispatching
economy
electricity industry
environmental sustainability
environmental values
equilibrium optimizer
evaluation platforms
failure
fate and transport
fdia
features selection
fire
fork
fuel consumption
fuzzy feature selection
fuzzy features
henan
heuristic algorithms

heuristics
hydraulic modeling
hydraulic structure
important features
increments
informatization constructions
lake pollution
landscape architecture
laws and legislation
leonardo
leonardo's rule
live-fire
load-balancing
load-bearing capacity
loading
loads
model fusion
model monitoring
nanyang
ocean currents
offshore oil well production
on-line state
one-dimensional
physical simulation
pipe model theory
platform
pollination
pollutant diffusion
pollutant transport
power generation
power internet of thing
power production
power sector
precise modeling
production control
real time observation
real time sensing
real time tracking
real-time prediction
recurrent neural network (recurrent neural network)
reliability improvement
risks assessments
river water quality
sdg 11
shape
short term load forecasting
simulation engine
smart city and smart grid

smart grid technologies
smoke prediction
social development
social management
standard tests
stormwater models
sustainable energy management
system framework
test scheme
texas
thermal dynamics
thermostatically controlled loads
tidal currents
tidal power
tracking and predictions
treatment plants
tree risk assessment
trees
urban fire
urban renewal
urban river
urban road planning
urban water systems
utility companies
utility sector
water quality data
water quality evaluation
waterworks
highly accurate
urban-rural
2-d model
3d point cloud data
5d model
backdoor attack
backdoors
bp neural networks
coal
coal and gas outburst
collaboration algorithms
computational efficiency
computer games
deep learning model
digital twin.
dynamic importances
dynamic priority
energy-efficient trajectory planning
full cycles
game engine

gas bursts
geometric digital twin
geometric feature extraction
global misclassification
grey relational analysis
indoor positioning systems
information transmission
intelligent collaboration algorithm
learning models
logic gates
mechanical
mechanical experiment
misclassifications
multi-level dynamic priority and importance scheduling algorithm multi-le
multilevels
navigation
object position shifting
object positions
objective functions
outburst intensity
partial misclassification
pipe networks
pipeline networks
point cloud data
position shifting
power grids
precipitation intensity
prediction algorithms
prediction methods
pumping plants
pumping stations
qualitative analysis
quantitative condition assessment
quantitative conditions
quantum gate node
quantum gates
quantum theory
renewable sources
sensor communication
specific object vanishing
stochastic loads
terrestrial mobile computing
trajectory planning
trigger
underground construction
underground pipeline
underground pipeline network
unnecessary object generation

urban pipe network
water energy
water energy management
water pipelines
wavelet embedding
web browsers
webgis
webgl
wind energy sources
wind turbine
urban health
visual analytics
cognition
finite element method
hydroelectric power
surface roughness
travel demand
adaptation models
archetype characterization
autonomous drone
behaviour models
big graph data
building fires
building mesh
carbon peaks
cellular telephones
charging station planning
clusterings
complex buildings
connectivity
cost-efficiency
critical fields
cybe-physic
cybe-physical social intelligent ecosystem
physical social intelligent ecosystem
data allocation
data engineering
differential privacies
digital twin modeling net-zero energy
energy landscape
epidemiology
experimental test
exposure
extreme hot weather
extreme weather
finite element modelling (fem)
fire extinguishers
foundation

foundation pit excavations
foundations
frequent subgraphs
gaming engines
gated recurrent unit
generative model
graph data
greenhouse gas emission reduction
health and safety
hierarchy algorithm
high dimensionality
highway engineering
histogram
home networks
hybrid power
hybrid renewable system
hydrograph
hydrographs
iber
image resolution
immersive visualization
insel4city
integrate renewable energies
isolated city
kriging surrogate model
latin hypercube sampling
mobile agents
mobile communication networks
mobile phone datum
model rainfall-runoff
modelling strategies
multi-criteria approach
multicriteria analysis
multigraphs
multilayer neural networks
network data
network technologies
numerical implementation
parametric finite element model
parametric finite elements
personal communication systems
photovoltaic cells
point completion
privacy regulation
pumped storage
rainfall-runoff modeling
regeneration model
renewable energy strategies

renewables
research focus
risk-informed decision making
roof
roof slopes
security and privacy
signal control
skip attention
smart city energy management
smart energies
smart firefighting
smart homes
smart system
social intelligence
social sciences computing
solver optimization algorithm
spatiotemporal information
strata deformations
supply chain management
surrogate method
surrogate modeling
sustainable charging
swmm
telecommunications networks
temperature
thermal exposure
traffic light managements
traffic management systems
travel demand modelling
tunnel deformation
two ways
underground structures
unreal engine
urban building energy modeling data preparation
urban heat island
urban planners
urban region
urban transportation systems
waste disposal
windmill
industrial ecology
manufacturing industries
public transport
3d urban modeling
analysis frameworks
arid regions
artificial intelligence learning
backscatter

bat algorithms
benchmark suites
benchmark testing
bim-gis-iot integration
bridge construction
bridge constructions
broad-angular coverage
brownian bridge
building codes
building community
building management
building simulation
carbon reduction
cause and effects
changing climate
civil structure
cobie
communicationtechnology
community building energy modeling
community buildings
complete system
connected vehicle
data edges
data envelopment analysis
data envelopment analysis models
data systems
data-driven and knowledge-driven
data-driven model
data-driven urbanism
delay
developing countries
development model
digital assets
digital economy
digital platforms
distributed generation system
dynamic scheduling
dynamics models
economics analysis
efficiency evaluation
energy autonomous
energy conservation building code
energy flow algorithms
energy hub system
energy saving and consumption reductions
equity
exponentials
farming system

generation method
geographical environment
gis integration
gps
green economy
green's expansion
health care 4.0
high quality
hub systems
indoor farming
infrastructure maintenance
instance segmentation
instrumentation
integrated maintenance
integrated operations
intelligent driver models
intelligent road inspection
intelligent roads
internet-of-thing swarm
iran
justice
key elements
landscape
landscape visualization
large dataset
linear transformations
linked datum
local coordinate system
logistics
long short-term memory.
low-carbon cities
malaysia
millimeter wave identification
millimeter wave lens
millimeter-wave identifications
model method
modified bat algorithm
monitoring and control
nonlinear programming
occlusion handling
occlusion problems
on demands
on-demand logistic
optical design
optimization framework
petroleum reservoir evaluation
philosophical aspects
pixels

planning scheme
platformization
practical deployment
primary objective
process dynamics
public transport systems
pump station
radar cross section
radar cross-sections
rami 4.0
rapid optimization
real-time information systems
research priorities
research work
residential areas
resource allocation and planning
reusability
rigid body transformation
road inspections
road pothole detection
rough surfaces
route planning
rural and urban
search algorithms
secure scheduling
semantic integration
semantic-web
smart manufacturing
smart transportation
social equity
social justice
spatial-temporal data
spatio-temporal data
storage facilities
street image
sustainable infrastructure
synthetic data
synthetic datasets
system reliability
systems architecture
tehran [iran]
tehran [tehran (prv)]
three-dimensional city model
timing
traffic flow prediction
training sets
transdisciplinary research
transport networks

transportation digital twin
transportation planning
tropics
twin modeling
urban 3d model
urban analysis
urban logistics
urban researches
urban spatial information
urban water supply system
various technologies
vehicle modelling
vehicle system
vehicle trajectories
vehicular dynamics
virtual geographical scene
visualization and analysis
water treatment plant
water-supply systems
web-based augmented realities
yolo
adaptive management
convergence of technologies
hamburg [germany]
urban dtc
3-d recovery
3d construction
3d scenes
3d web interface
allocation model
application integration
architectural visualization
architecture technology
asset maintenances
autonomous car
base stations
bayesian inference
bertopic
big data analyse
biomimetics
bridge networks
building certificates
business intelligence
byzantine agreement
centrality
channel estimation.
civic architecture
civil society

co-production
cognitive graph
computation theory
computational analysis
computer aided analysis
computer aided design tools
computer circuits
computer software maintenance
computer technology
computer-aided design tool application at city scale
computing platform
computing technology
condition monitoring
cooling systems
cpo
critical response
cutting edges
data acquisition system
data analysis models
data analysis system
data-driven optimization
data-transmission
digital elements
digital twin deployment
digital twin integration
digitisation
dimensional representation
distributed parallelism
dlt
dynamic channels
edge services
educational institutions
electric vehicle supply equipments
electron holography
ethiopia
fault detection and diagnose
fault detection and diagnosis
federated byzantine agreement
flood management
flood management digital twin platform
flood mitigation
flood propagation
floor
gas detectors
gas pipe networks
gaussian distribution
gaussian noise (electronic)
gaussian splatting

gaussians
geolocations
geospatial observation
graph deep learning
green & smart city
ground control points
hadar
heterogeneous data integrations
hierarchical systems
holographic displays
http
human analysis
human pose
image enhancement
information platform
intelligence
internet of thing/internet of everything
its applications
key issues
latent dirichlet allocation
latent dirichlet allocation model
learning approach
lightemitting diode
lighting efficiency
low power
luminance
management of structures
massive data
mechanism-based
mesh repair
mesh simplifications
microservice
middleware
mirror worlds
mirrors
multi-level mesh
multihop transmission
multiscale modeling
multisensor integration
multisource data
net zero energy
node-red
novel architecture
offline
on-line gaming
ontologies and rules
operation rules
parallelism strategies

partial shading
physical elements
physical locations
physital space
prediction performance
probabilistic graphs
probability distributions
ptz camera
public streets
pump operation
pump operation rule
pumping station operation
quality-preserving
radiation analysis
ray tracing
real time data acquisition
real time kinematic
real time traffics
real time videos
recovery
remote location
remote sensing system
repair
reputation mechanism
road network
safe driving
safety engineering
safety performance
scientific literature
security performance
security systems
sensor technologies
sensor web
server sides
server-side business logic
smart city architecture
smart city workflow
smart lightings
smart stormwater management
solar energy harvesting
solar radiation analyse
spatial cognition
spatial perception
splatting
storm sewers
stormwater managements
street lamps
street lighting

structure systems
structured analysis
switching matrix
system applications
system digital twin
system prediction
system safety
tall buildings
time-critical
tool applications
topic modeling
transportation network
trustless service
urban governance
urban government
urban visualizations
user need
user-generated
user-generated content
variation bayesian inference
vehicular environments
video cameras
video monitoring systems
video registration
video-streaming
virtual geographic environments
visual analysis
visual perception
visual programming
visual similarity
visualization technologies
web interface
web-sites
workflow
artificial intelligence techniques
commerce
public space
accuracy
adaptive algorithms
aerial images
agent framework
aggregates
anonymization
anpr camera
ansys hfss sbr+
antennas measurement
application programs
arts computing

auto-regressive
automated object detection
automated workflow
autoregressive integrated moving average
avatar
categorization systems
close range photogrammetry
cloud classification
community values
competition
competitive situation
computing abilities
continuous improvements
count datum
customer as a service provider
customer expectation
cycle path
data capability
data organization
derived information framework
development trends
distribution systems
dynamic wireless networks
economic models
edge detection
electric power utilization
environmental behavior
epanet
error correction
evolutionary trend
facebook
facebook prophet
geometrical optics
geospatial visualization
global coordinates
governance
graph model
hidden markov models
hidden-markov models
high-dimensional
high-dimensional indicator
higher-dimensional
human society
hydraulic tools
image edge detection
information framework
information modeling frameworks
integer coding methods

interactive computer graphics
isovist field
isovists
its
jointly spatial-temporal analyse
knowledge capabilities
knowledge gaps
ladders
landscape elements
laser scanning
lidar integration
logistics service provider
loss measurement
low-emission transport
macau
machine learning models
management and controls
manufacturing sector
map
market research
market researches
mathematical transformations
maturity model
maximum likelihood
media streaming
microwave antennas
millimeterwave communications
mobile laser scanning
model generation
monte carlo methods
movement
moving averages
multi-object tracking
multiple object tracking
network tools
network transmission
nist
object detection systems
open datum
outdoor comforts
patent
patent analysis
patents and inventions
pedestrian movement
penetration loss
petaflops
petascale
photo-realistic

photosynthetically active radiation
planning management
point cloud classification
potential sources
privacy issue
privacy preservation
procedural generation
propagation
propagation loss
public clouds
public participation
public transit
random variables
rapid growth
rapid transit
raytracing
real road networks
real-time data processing
real-time object detection system
realtime traffic
rigorous evaluation
scada systems
second life
semantic agents
semantic interoperability
sensitive datas
service provider
sewage system
shape measurements
signal acquisitions
situation awareness
slam
smart phones
smart urban water network
software frameworks
spatial data analysis
spatial dimension
spatial relationships
spatial temporal analysis
spatial vitality
statistical datas
streaming medium
supercomputers
system scaling
tachometers
technical-social framework
terragraph
three dimensional space

three-dimensional digital models
three-dimensional grids
three-dimensional model
time series forecasts
traffic analysis
traffic signal controllers
traffic simulators
transformation parameters
trees (mathematics)
tuning
turning movement count
urban delivery
urban simulations
urban studies
urban traffic networks
urban waters
value engineering
vegetation
vehicle license plates
vehicle performance
vehicles managements
vehicular ad hoc networks
vehicular adhoc network
virtual replica
vision model
visual exposure
voxelization
wages
wait time
walking
water distributions
water network tool for resilience
weather forecasting
web 3.0
welfare impact
wi-fi
data
access control
adaptive multi-objective multi-task service composition
athens [attica]
attention level
authorization
battery energy storage
benchmark datasets
biological system modeling
breakings
building construction
building extraction

building life cycle
building lifecycle data
building permits
building process
building structure
building systems
citizen report
cloud manufacturing
cogeneration and trigeneration
communication layers
construction efficiency
construction settlement law
coronaviruses
current practices
cutting
cutting machines (mining)
cutting model
cybe-physical-social ecosystem
physical-social ecosystem
damage simulation
data sharing
data visualisation methods
data-centric model
data-driven decision-making
data-driven society
dbscan
decoding
developing world
development process
different stages
digital universalism
disaster prevention monitoring system
disease control
district heating networks
drone transportation system
dynamic clustering
dynamic controls
earthquake damage simulation
earthquake damages
earthquake disaster
earthquake emergency rescue
earthquakes
edge clouds
edge-cloud computing
edge-to-cloud
elastoplasticity
electric batteries
electric load flow

emergency rescue
emission dispersion
emission model
encoder-decoder
energy consumption model
energy demands
energy metabolism
england
epidemic simulations
equipment monitoring
experimentation
faulting
fog manufacturing
forecasting models
fully integrated
general architectures
geobim
geodataframe
graph-based
graphical method
head frames
health-care managements
heating equipment
heterogeneous intensity
hospital information systems
hospitals
hourly forecasting
household loads
human settlement
hydroponic farm
hyper-parameter
imaging equipment
inadequate data
industrial iot
industrial loads
information coordination
information integration
information model-based urban planning
information models
information retrieval
integrated circuits
integrated emissions
intelligent planning
interruptible load
interruptible load scheduling
inverse problems
inverse reinforcement learning
large scale internet

lazio
lighting
load scheduling
london [england]
management challenges
metabolism
microgrids
mobility justice
model informations
multi objective
multi tasks
multimodal data
multiple technology
multiscale visualization
natural disasters
nft
novel coronavirus disease
observatories
on currents
on-currents
one stop shops
overall planning
overall planning of underground space
patient trajectories
peak demand
per unit
photovoltaic panels
piles
planning models
practical constraint
prediction accuracy
prevention and controls
privacy
privacy awareness
process structures
process system
public affairs
python
real time analysis
real world projects
real-scene 3d model
real-space
risk analysis
road condition
roadbuilding machinery
roadheaders
robot sensing system
roma [lazio]

rome
seismic response
services composition
signal encoding
smart technology
social aspects
social ecosystems
social entities
social value orientations
socio-technical perspective
software agents
soils
spatial data fusion
subsidy system
surveillance sensors
systematic
thermal insulation
three-dimensional reality
timeliness/accuracy
traffic guidance
tri-generation
trnsys
underground equipment
underground farm
underground space
university sector
urban attention
urban district clean energy transition
urban emergency management
urban feature extraction
urban features
urban regions
urban scale digital twin
urban-integrated farm
user subsidy rate
video fusion
vr modeling
windows operating system
working environment
high-accuracy
urban society
3-d printing
3d printing
3d urban aerial network
3d-printing
additives
advanced rail transit
aerial networks

agricultural purpose
agriculture management
animals
antenna grounds
array devices
automated decision making
automotive industry
bi-directional communication
biodiversity
biodiversity conservation
building efficiency
building production
camera trap
classifieds
climate change projections
climate models
cloud-based
cloud-fog-edge computing
fog-edge computing
coactivity
collaborative health care
commercial building
commercial vehicles
communication technologies
communications modeling
conservation
construction machinery
construction process
continuous-flow
contrastive learning
control-flow
coordination reactions
crop farming
cyber threats
cyber-physical systems (cps)
cyberthreats
damage detection
data explosion
data trustworthiness
data-communication
debris flow
detection system
digital management
digital shadow
distributed ledg technology
driverless
driverless vehicle
dynamic links

efficient learning
electric measuring instruments
electric power measurement
energy benchmarking
energy generations
energy resources
energy sources and sinks
energy-efficient federated learning
enhanced learning
enriched semantic segmentation
environmental consequences
environmental footprints
farms
fault tolerant communications
flexible electronics
flight simulators
flood damage
fog layer
future energies
future research directions
geographic database
geographical information
geospatial information
greenhouse management
ground systems
growing demand
guardian
handhelds
handhold lidar system
haptics
helicopter services
high reliability
high speed rail
human actions
human-on-the-loop
human-vehicle coactivity
humanin-the-loop
hydrogeology
image
information networks
ink-jet printing
intelligence integration
intelligent technology
internet protocols
interspecies money
intrusion-detection
knowledge representation
label-efficient learning

landslide
learning capabilities
lenses
low-cost lidar sensor
low-high
machine learning approaches
machinery
maglev train
magnetic levitation
making decision
management methodologies
management practice
manufacturing techniques
mean-field
mean-field reinforcement learning
metals
mixed-reality environment
mobile 3d
mobile 3d mapping
natural disaster
natural hazard
networks learning
new solutions
ngmn
object systems
origami
partial data
patient health
phased-arrays
physical process
planning practice
ports and harbors
pre-training
printed materials
prior-knowledge
prior-level fusion
process management
production line
production optimization
project schedules
proposed architectures
public safety
rail transit
rail transit systems
railroad transportation
real-estates
real-time data
responsible artificial intelligence

reward systems
road users
salient features
schedule delay
seaport
security issues
self drivings
self-supervised
self-supervised learning
semantic knowledge
semiconductor devices
semiconductor materials
simplified analysis
simulation of buildings
smart agricultures
smart greenhouse
smart meters
smart port
smart public safety surveillance
solid waste
solid waste management
south africa
supervised learning
tactile internet
task-oriented
technological advances
temporal fluctuation
traffic spatial information network
transport infrastructure
transport solution
trucks
unified communications
unlabeled data
urban data mining
urban modelling
users infrastructure
vehicle routing
vehicle routing problem
vehicle routing problems
vertiport
virtual models
volunteered geographical information
vulnerable road user infrastructure mapping
waste production
waste to energy
wildlife populations
workflow management
world population

zero day attack
zero-day attack
zero-day attack detection
cloud platforms
data analysis
engineering
integrated platform
parameterization
68t37-reasoning under uncertainty in the context of artificial intelligence 68t37-reasoning unc
afforestation
air grounds
air-to-air collisions
aircraft accidents
alignment
amplitude modulation
approximation algorithms
arch bridges
architectural construction
architectural engineering
architectural engineering constructions
architectural innovation
attention mechanisms
automated solutions
automated vehicles
autonomous unmanned aerial vehicles
awareness
beam alignments
bicycles
bim
biophilia
biotechnology
centralised
cloud integrations
cloud services
coastal barrier
cognitive digital twin
cognitive technology
collaborative robots
combinatorial optimization
combined heat
combined heat-power
common environment
complex physical systems
construction 4.0
construction management
construction operations
construction sites
context-aware interaction

coordination and control
correction models
data algorithm
data assets
data representations
data-driven analysis
decarbonization
decentralized semantics
descriptive data-driven analyse
design frameworks
detection and localization
digital technology
digital twin state updating
disaster relief
disaster response
dynamic applications
economic feasibilities
edge-device-cloud integration
device-cloud integration
electric excitation
electrical power
electro mobilities
engineering management
envi-meet simulation
environmental impact assessment
error prones
ethical technology
facade materials
facades segmentations
fair allocation
faults diagnosis
flying objects
framework development
freight parking management
functional sites
future internet
gardens
gateways (computer networks)
gender issues
generalized assignment problem
geographical scale
glasgow [glasgow (ads)]
glasgow [scotland]
hazard prevention
hazards
heat power
heterogeneous data
heterogeneous environments

heterogeneous networks
high-level systems
highway bridges
hub motors
human experiment
hurricanes
indirect communication
individual modeling
influencing mechanisms
information elements
integrated frameworks
integrated networks
integrated sensing
integrated sensing and communication
integration issues
intensity
intensity features
internet of things industries
kernel density estimate
knowledge intensive tasks
last mile delivery
leak localization
leaks detections
levees
life cycle stages
life-cycle management
lifecycle
lightweight digital twin
lightweighting
link scheduling
load capacity
load identification
load ratings
localized events
location information
machine learning techniques
management problems
meta smart twin city
minimum cost
minimum-cost generalized assignment problem
mobile ad hoc networking
mobile ad hoc networks
mobile ad-hoc networks
model evolution
modeling simulation
multi-machine collaboration
multi-machines
multi-unmanned system

multipurpose robots
nasa
natural gas pipelines
need analysis
network-based analysis
noise abatement
ontology and semantic
operation management
operation planning
optimal controls
optimal decisions
optimized production technology
overall efficiency
parking management
payload management
pipeline fault diagnose
pipeline leakage
pipeline leaks
pixel scale factor
power-train systems
powertrains
product life cycle management
property damage
radiometrics
radiometry
railroads
railway platforms
railway stations
randla-net
reasoning under uncertainty
remote monitoring
research interests
research method
research questions
research results
research trends
reservoirs (water)
road bridge
road fixture
road profile
scale factor
scheduling algorithms
scotland
sea level rise
semantic communication
semantic data
service modeling
service oriented architecture (soa)

services and applications
services network
setup costs
smart energy systems
soa (serviceoriented architecture)
social internet of thing
socio-technical concern
spatial impacts
state refreshment of service model
strategy development
surface materials
symbol
synchronization signals
system models
system simulations
t-cells
telecommunications traffic
tera hertz
terahertz waves
testing environment
time-frequency analysis
toll bridges
tracked vehicles
tradeoff between the state freshness of service model and the update cost tradeoff between
traditional settlement
traffic management centers
traffic prediction
transport
ubiquitous semantic metaverse
unauthorized access
unmanned system
urban district
urban domains
urban ecosystem
urban modeling
urban safety
user's preferences
vector control (electric machinery)
vehicle load
vehicle mission
verification-and-validation
video data
dna
questionnaire survey
3d building models
active appearance models
adaptive gravitational search algorithms
aerial surveys

air communications
air space
air taxi
air transportation
aircraft maintenance
aircraft systems
airspace management
ambulance
ambulances
and execution
annual operating costs
anthropogenic heat
anthropogenic heat release
anthropogenic heat transfer
application management
artificial intelligent
as-build modeling
basic concepts
big data analytic
bim models
bioinformatics
building monitoring
cellular technologies
chassis dynamometers
classification approach
classification methods
cognitive overload
cognitive systems
collaborative decision making
company operation
computational platforms
computer debugging
connected autonomous vehicle
converting industries
cooling loss
curb environment
daily lives
darknets
data enhancement
data framework
data intensive
data ownership
data-rate
decision-making process
deforestation
deterministics
digital twin maturity
digital twin technology evolution

dna model
dna sequences
driving behaviour
driving cycle
dynamic adjustment
dynamometers
early warning
edge networks
efficiency and reliability
electromagnetic environments
energy productions
environmentally friendly alternatives
evolution direction
experimental platform
explosive growth
eye protection
feasible solution
functional units
geography markup language
geometric filters
geovisualization
gis data
glaucoma
global navigation satellite systems
google+
grape dataset
ground truth
heat flux
heat transfer
high speed
human bodies
human subject experiments
implementation layer
individual differences
industrial operations
informatization
infrastructure construction
innovative solutions
integral view
integration frameworks
intelligent information systems
intrinsic differences
investment costs
legacy systems
license
long-term decisions
loss rates
low altitudes

maintenance management
mapping data
mass transportation
medical digital twin
metro services
metro system
metro tunnel
mobile users
mobility service
modeling reconstruction
modelling time
multi systems
multi-layers
multiagent deep deterministic policy gradient
multiscale atmospheric modeling
nadir images
national infrastructure
network-based
neuroimaging
new services
object verification
on-site assembly
operational efficiencies
operational modeling
ophthalmologic diagnose
ophthalmology
organ systems
path plan
personalized information systems
physical communications
planning and scheduling
plant shutdowns
point clusters
point features
policy gradient
policy makers
port
port city
power output
pre-cast
product optimization
propagation environment
quality evaluation
quebec city
radio
radio communication
radio maps
random forests

ray launching
recognition accuracy
reducing costs
regional energy systems
research communities
roadside units
root mean square errors
satellite signals
satellite visibility
selection
self-verification
semantics mappings
service matching
service platforms
services applications
session key
sign recognization
single-shot
site assembly
small target detection
smart city experimentation
smart vehicles
spatial scale
spatiotemporal intelligence
street imagery
structural performance
subways
support vectors machine
surface albedo
sustainable transport
system theory
systems operation
systems research
technological innovation
technology element
technology evolution
test bed
thematic information
time connectivity
time-consuming tasks
track modeling
traction forces
traffic intensity
traffic safety
trust mechanism
trusted computing
tunnel monitoring
tunnels

uas
uncrewed aircraft system
unified modeling
urban canyons
urban heat
urban infrastructure construction
urban planning managements
urban process
urban thermal environment
user associations
vehicle technology
verifiability
video technologies
virtual cities
virtual objects
virtual reality technology
visibility estimation
access points
accumulated damage
adaptive capabilities
adaptive manufacturing
aging population
applications domains
artificial intelligence tools
asset management
assets management
auction.
augmented-mixed reality
automotive systems
bibliometric study
bill of materials
biosensors
boundary models
building roof
building rooftop classification
c (programming language)
capital
carrier mobility
cesium platform
classification system
closed boundary
cloud computing platforms
co-creation
complex model
complex software systems
complexity
comprehensive analysis
computational model

computer forensics
computing devices
consensus mechanism
context-aware systems
core functions
costs consideration
critical evaluation
data modeling
data streams
decarbonising
decarbonizing city
decentralized networking and computing
design evaluation
design specification
design-science researches
detection process
digital forensic framework
digital world
disaster prevention and mitigations
disaster resiliences
disaster simulation
distribution transformer
distribution transformer sensing
domain experts
domain specific
double auction
drinking water
driver's safety
eco-friendly
elderly populations
electric transformers
electric utilities
electromagnetics
electronic crime countermeasures
em skin
emergency
emergency response information systems
emerging trends
energy infrastructures
engineering applications
environmental parameter
european test feeder
existing systems
expenditure
experimental approaches
extracting information
factors analysis
fire department

footprint identification
forensic investigation
future mobility
game
global optimisation
global optimization
hardware and software
heatmaps
high-fidelity modeling
hot spot
hotspot temperature
hottest-spot temperatures
human drivers
idaho
idaho national laboratory
improve performance
in situ survey
individual mobility
industrial grids
information exchanges
information system design
inter-domain
iot integration
key technologies
know-that
knowledge transfer
landforms
large-scale systems
living conditions
living quality
low resource
low resource devices
low voltage distribution system
low voltage measurement
low voltages
manufacturing components
map planning
matchings
metaverse forensic
metaverse forensic framework
mining autonomous driving
mobile trajectory anomaly
mobility datum
modelling framework
modular framework
multimodal sensor
national security
natural environments

number of datum
nutrient exchange
onboard computers
orthophoto processing
orthophotos
overhead transformer monitoring
p2p network
parallel mine world
parallel testing
passive objects
pedestrian safety
performance metrices
personalized service
personnel training
physiological models
physiological signals
physiology
pilot programs
predictive capabilities
private vehicles
problems and challenges
procedural content generations
processing speed
processor scheduling
project case
public administration
radio transceivers
real test feeder
real world experiment
real-world implementation
recognition process
reconfigurable
recreation centers
reliability
remote sensing data
resilience strategy
resources security
reviews
roof dataset preparation
rudimentary level
rural grid
scientometrics
security surveillance
self control systems
service industry
site inspections
sky view factor
smart car

smart city development
smart city modeling
smart em environment
social issue and urban resilience strategy
social issues
software-systems
soil conservation
stadiums
statistical techniques
surveillance
synthetic tests
system behaviors
system resources
system-by-design
systematic mapping studies
techniques and tools
technology platforms
technology transfer
temperature and humidities
terrain generations
test feeders
testbeds
testing platforms
theoretical framework
three-dimensional topography
tool support
traffic safety measures
traffic systems
transformer monitoring
transportation and urban planning
tree canopy
underlying systems
urban
urban agricultures
urban disaster
urban energy modeling
urban grid
urban settings
use-case
user centered design
users' acceptance
valencia [comunidad valencia]
valencia [valencia (prv)]
vehicle safety
vehicle-based systems
video
virtual/augmented/mixed reality
virtualization technologies

voltage distribution measurement
waters resources
wearable technology
wet weather
wireless communication links
wireless network planning
greenhouse gas emissions
abundance
alert systems
anomaly classification
bidirectional model transformation
bidirectional transformation
biocommunications
building materials
building owners
building scale
business case
c-ran
camera controls
car-as-a-service
cleaning
comfort
commercialisation
customer experience
d-band
data challenges
data traffic
data-source
decision method
delivery service
delivery time
densifications
design and implementations
design approaches
design domains
design strategies
digital objects
dresden
electron diffraction
emissions reduction
feasibility studies
flow time series
form factors
freeway systems
gate control
generative design
geo-spatial data visualizations
geospatial applications

granularity
greenhouses gas
hazardous weather condition
heat transition
human behaviors
indoor mobility
industry 40
infrastructure assets
it focus
kyoto protocol
line of sight
lines-of-sight
loop detector
machine-readable format
mfd
mobility solutions
model based development
model-driven engineering
multiple heterogeneous robots
multiple-robots
municipal utility
national laboratory
network deployment
new business models
office workers
open interface
open platforms
output variability
pedestrian routes
photovoltaic potentials
process automation
programming technique
qgis
ramp metering
re-farming
real time interactions
robot control technologies
robot process automation
robots
sales
saxony
semantic information
sensitivity analysis
sensors faults
simulation and optimization
singapore
smart city semantic
smart freeway

software engineering
solar irradiance
solar radiation model
specific energy
spectra's
stand structure
standard design
the concept of digital twin
the synergy of digital twin
time series
traffic flow time series
traffic modeling
traffic-shaping
twin design
urban analytic
urban energy planning
usage scenarios
watershed
weather data
web based
web-based information
wireless fronthaul
actor critic
administrative system
air traffic management
application contexts
architectural modeling
architectural models
athens [georgia]
augmented and virtual realities
automated coupling
automated data
belief systems
cellulars
city centers
collaborative platform
commercial applications
concept definitions
demand data
development of science and technologies
digital divide
digital radio
discretization
disparate systems
e-sciences
effective management
emergency traffic control
essential characteristic

evacuation time
evtol
exploring engine for the future society
externality
fine granularity
flight management
flight operation
free communications
frequency estimation
frequency sensors
georgia
global connectivity
guangdong
guangzhou
heat island
high tech industry
hotspots
human activities
illinois
industrial production
interactive technology
it applications
large scale simulations
medicine
metasurface
mining
modernization
multi-modal information
multidimensional information
multifunctionals
natural systems
new age
new south wales
new york
novi sad
operational procedures
operational research
optical depth
pedestrian simulation
pedestrian traffic
pedestrian traffic simulation
personalized medicines
physical parameters
policy evaluation
policy-based
power sources
research roadmap
road traffic flows

scheduling policies
sensing networks
sensing scheduling
serbia
simulation practices
single-frequency
soft actor-critic
software architecture model
summer
sustainable digital transformation
sydney [new south wales]
time pressure
traffic flow simulation
traffic volume data
universal mobile telecommunications system
urban air mobility air traffic management
urban air mobility integrated flight management
winter
3d spatial infor mation
absolute position
android platforms
apartment buildings
artificial agents
authoring tool
automobiles
bologna [bologna (ads)]
bologna [emilia-romagna]
calculation models
chonju
collaboration and interaction awareness
comparative assessment
compliant devices
dashboard
design and construction
digital information
digital-twin-integrated platform
emilia-romagna
empowerment
energy performance
environment sensing
event-driven
framework architecture
gaining insights
health
healthcare services
heating demand
human agent
human being

individual-based
industrial processs
infrastructure facility
infrastructure networks
infrastructure providers
integrated app
intelligent agents
intelligent mobility
interface states
intergovernmental panel on climate change
learning curves
meteorological variables
mobile applications
mobile systems
mobility system digital twin
model requirements
multiple services
non-trivial
north cholla
objects recognition
performance modeling
personal health devices
physical mobility
position information
primary energy demand
shared spaces
smart mobility
social well-being
switzerland
trade off
transport systems
urban climate
urban heat environment
urban heat monitoring
visual languages
well being
widget
zurich [switzerland]
zurich [zurich (ads)]
3d deep learning
advertizing
architecture and urbanism
architecture engineering
architecture-based
billboard exposure analyze
biological entities
bottom-up approach
building energy consumption

building information physical model
carbon fiber
community management
composite material
computer simulation modeling
critical factors
data architectures
delphi
digital practice
digital twin framework
electric power supplies
energy forecasting
essential elements
eurasia
exposure analysis
forecasting system
formal modeling
formal validation
future study
general systems
generic architecture
glass fiber
human mobility
human mobility data
human-centric application
individual components
infrastructure policy
interactive system
investment
learning-based methods
linked open datum
marketing tools
matrix
model drive
modeling and validation
north america
npc
organisational
piraeus
planning process
planning systems
power supply
primary contribution
railway points
railway scene
roadmap
rural area
self monitoring

semantic map
skin
social practices
socio-ecological systems
theoretical modeling
urban economy
user-centered digital twin
user-centred
viral disease
virginia
visual analytics systems
wind
aircraft
alabama
capital city
defense in depth strategy
diagnostic accuracy
dublin [(cty) leinster]
florida [united states]
height
indonesia
interdisciplinary approach
irbid
ireland
israel
leinster
levee
mobile
model learning
privacy enhancing technologies
privacy preserving
privacy-preserving techniques
psychology
rainwater
refugee
seoul [seoul (ads)]
seoul [south korea]
terrain
action plan
biofuel
citizen engagements
data usage
delta
extraction method
hamburg [hamburg (stt)]
health worker
new zealand
nigeria

optical and wireless
pearl river [united states]
regional development
regional economy
rotterdam
rural development
rural economy
rural planning
signal processing
south holland
spatial variation
spectral analysis
state of research
temporal variation
urban-rural migration
zhujiang delta
analytical framework
and internet of thing
cartography
freight transport
gartner
grassroots level
land management
land market
land tenure
nonlinearity
pakistan
scientific knowledge
systems implementation
technological trends
alternative agriculture
curriculum
education
exploration
fine grained sediment
gas industry
laser method
singapore [southeast asia]
student
teaching
automobile industry
classification
computer networks
diffusion
imaging method
language
methodology
phase transition

regeneration
semantic standardization

	degree
smart city	1.000000000
decision making	0.542478566
digital twin	0.491816056
internet of things	0.473499610
urban area	0.446609509
urban planning	0.382696804
information management	0.321901793
sustainable development	0.312159002
deep learning	0.310600156
e-learning	0.293842557
algorithm	0.266173032
three-dimensional modeling	0.251363991
artificial intelligence	0.250974279
energy efficiency	0.245518316
three dimensional computer graphics	0.245128605
machine learning	0.243959470
architectural design	0.227591582
virtual reality	0.226812159
article	0.226032736
optimization	0.225643024
embedded systems	0.201870616
real time systems	0.198752923
sustainability	0.197973500
semantics	0.195245518
energy management	0.194466095
3d modeling	0.194466095
urban development	0.187840998
urban transportation	0.180826189
alternative energy	0.180046765
climate change	0.176539361
network security	0.174201091
antennas	0.172642245
building information modelling	0.172642245
urban growth	0.165627436
learning systems	0.164848012
life cycle	0.164458301
gis	0.163289166
urban areas	0.156274357
visualization	0.153936087
energy utilization	0.153156664
building	0.144193297
forecasting	0.141075604
digital storage	0.139906469
human	0.139906469
block-chain	0.136788776

data visualization	0.133671083
interoperability	0.130943102
data handling	0.130553390
blockchain	0.130163679
decision support systems	0.124707716
machine-learning	0.123928293
unmanned aerial vehicles (uav)	0.123148870
smart power grids	0.122369447
china	0.121979735
data driven	0.120031177
cyber physical system	0.117303196
computer simulation	0.116913484
information theory	0.115744349
real- time	0.114964926
real-world	0.114185503
smart grid	0.113795791
intelligent systems	0.113406080
risk assessment	0.109898675
interactive computer systems	0.109898675
big data	0.108729540
data acquisition	0.108339829
real - time system	0.106780982
security	0.106391270
geographic information systems	0.106391270
digitization	0.106001559
urbanization	0.105611847
urban environments	0.103663289
numerical model	0.102104443
optical radar	0.100155885
urban design	0.098207327
point-clouds	0.098207327
mapping	0.098207327
case-studies	0.097427903
aerial vehicle	0.097038192
city model	0.095089634
disasters	0.094699922
water management	0.094310210
data set	0.093920499
metaverses	0.092751364
floods	0.091971941
vehicles	0.091582229
large-scales	0.091582229
optimisations	0.090802806
edge computing	0.090023383
built environment	0.089243959
artificial neural network	0.089243959
data integration	0.088074825
urban digital twin	0.087685113

roads and streets	0.087685113
construction industry	0.087295401
intelligent buildings	0.086515978
water supply	0.086126267
object detection	0.085736555
5g mobile communication systems	0.085346843
buildings	0.084957132
land use	0.084567420
controlled study	0.084177708
remote sensing	0.083398285
computational modelling	0.082229150
adult	0.082229150
behavioral research	0.081839439
unmanned aerial vehicle	0.081839439
automation	0.081449727
modeling	0.079890881
cost effectiveness	0.079890881
renewable energies	0.079501169
uncertainty	0.079111458
quantitative analysis	0.079111458
ontology	0.078332034
strategic approach	0.077942323
sensor	0.077552611
intelligent transportation systems	0.077552611
learning algorithms	0.077552611
spatial data	0.077552611
economic and social effects	0.077552611
data analytics	0.077162899
simulation	0.077162899
traffic congestion	0.076773188
stakeholder	0.076383476
geo-spatial	0.075214341
internet	0.073265783
3d-modeling	0.072876072
flood control	0.072876072
traffic management	0.072486360
real time	0.071706937
software	0.071317225
energy	0.070927514
long short-term memory	0.070537802
economics	0.070537802
urban transport	0.070537802
population statistics	0.070148090
benchmarking	0.069368667
'current	0.068978956
monitoring	0.068978956
semantic segmentation	0.068199532
floodings	0.067809821

autonomous vehicles	0.067420109
metadata	0.067030398
urban drainage	0.067030398
stochastic systems	0.066640686
sustainable development goal	0.066250974
3d models	0.066250974
technological development	0.065861263
united states	0.065861263
city	0.064692128
road	0.064302416
urban buildings	0.064302416
renewable energy	0.064302416
humans	0.064302416
proof of concept	0.063912705
emission control	0.063912705
augmented reality	0.063133281
australia	0.063133281
decision support system	0.062743570
carbon dioxide	0.062743570
comparative study	0.062743570
conceptual framework	0.062353858
resource allocation	0.062353858
costs	0.062353858
data mining	0.062353858
physical objects	0.061964147
office buildings	0.060795012
urban drainage systems	0.060795012
risk management	0.060405300
information modeling	0.060015588
ecosystems	0.060015588
experimental study	0.060015588
innovation	0.059625877
water quality	0.059625877
microgrid	0.059236165
drainage	0.058846454
mean square error	0.058456742
real time control	0.058456742
urban management	0.057677319
motor transportation	0.056897896
environmental technology	0.056897896
industry 4.0	0.056508184
digital devices	0.056508184
intelligent vehicle highway systems	0.055339049
network architecture	0.055339049
construction	0.055339049
database	0.055339049
performance	0.054949337
image reconstruction	0.054949337

transportation system	0.054559626
deep neural networks	0.054169914
solar energy	0.054169914
flooding	0.053780203
prediction	0.053780203
sewer	0.053390491
bayesian networks	0.053000779
computer crime	0.052611068
runoff	0.052611068
air quality	0.052221356
intelligent transportation	0.052221356
spatial analysis	0.051831645
local participation	0.051831645
environmental management	0.051831645
hydrodynamics	0.051831645
cost benefit analysis	0.051441933
data fusion	0.051441933
graph neural networks	0.051052221
virtual spaces	0.050662510
power	0.050662510
data processing	0.049883087
lidar	0.049493375
wireless communications	0.049493375
particle swarm optimization (pso)	0.049103663
urban energy	0.049103663
three-dimensional display	0.048713952
engines	0.048713952
cyber-physical systems	0.048324240
storms	0.048324240
energy use	0.047934528
iterative methods	0.047934528
pipelines	0.047934528
optimization algorithms	0.047544817
cyber security	0.047544817
energy management systems	0.047155105
melbourne	0.047155105
victoria [australia]	0.047155105
disaster prevention	0.046765394
reinforcement learning	0.046765394
cybersecurity	0.046765394
retrofitting	0.046765394
internet of thing	0.045985970
street traffic control	0.045985970
multi agent systems	0.045596259
building energy model	0.045596259
cyber physicals	0.045596259
integration	0.045596259
pedestrian	0.045596259

city scale	0.045206547
rain	0.045206547
uncertainty analysis	0.044816836
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italy	0.044037412
three dimensional displays	0.044037412
multiobjective optimization	0.044037412
technology adoption	0.044037412
demand-side management	0.043647701
virtual representations	0.043257989
performance assessment	0.043257989
urban systems	0.042868277
model predictive control	0.042868277
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intrusion detection	0.042478566
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detection method	0.041309431
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hydrological modeling	0.041309431
city digital twin	0.040919719
3d visualization	0.040919719
crime	0.040919719
digital replicas	0.040530008
historic preservation	0.040530008
cybe-physical systems	0.040140296
mapping method	0.040140296
neural networks	0.040140296
immersive	0.039750585
intelligent transportation system	0.039750585
3d reconstruction	0.039750585
solar radiation	0.039750585
energy model	0.039360873
solid modelling	0.039360873
computer vision	0.038971161
iot	0.038971161
3d city model	0.038971161
convolutional neural network	0.038971161
information system	0.038971161
enabling technologies	0.038581450
genetic algorithms	0.038581450
housing	0.038581450
data collection	0.038191738
sensors data	0.038191738
geographic information system	0.038191738
graphic methods	0.038191738

real-time control	0.038191738
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spatiotemporal analysis	0.037802027
building information modeling	0.037802027
data privacy	0.037802027
internet of things (iot)	0.037802027
future prospect	0.037412315
physical systems	0.037412315
autonomous driving	0.037412315
autonomous agents	0.037412315
work-flows	0.037412315
infrastructure	0.037022603
cloud-computing	0.037022603
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information and communication technology	0.037022603
mobile communications	0.037022603
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building model	0.036632892
large datasets	0.036632892
multi-objectives optimization	0.036632892
open source software	0.036632892
computer architecture	0.036243180
decisions makings	0.036243180
energy-consumption	0.036243180
industrial research	0.036243180
carbon emissions	0.036243180
environmental monitoring	0.036243180
pipeline	0.036243180
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mobile telecommunication systems	0.035853468
reliability analysis	0.035853468
resource management	0.035853468
roofs	0.035853468
time series analysis	0.035853468
errors	0.035853468
urban drainage model	0.035853468
water supply systems	0.035853468
sewage	0.035463757
5g	0.035463757
carbon emission	0.035463757
electric power distribution	0.035074045
knowledge graph	0.035074045
resources allocation	0.035074045
technological advancement	0.035074045
urban air	0.034684334

calibration	0.034684334
carbon	0.034684334
digital technologies	0.034294622
% reductions	0.034294622
drainage network	0.034294622
crowdsourcing	0.033904910
construction equipment	0.033904910
job analysis	0.033904910
systematic review	0.033904910
task analysis	0.033904910
water	0.033904910
geometry	0.033904910
hydrology	0.033904910
water levels	0.033904910
digital transformation	0.033515199
implementation process	0.033515199
adaptive control systems	0.033515199
building information model	0.033515199
building information model - bim	0.033515199
economic development	0.033515199
efficiency	0.033515199
energy conservation	0.033515199
twin networks	0.033515199
technology	0.033125487
digital twin city	0.033125487
internet of vehicle	0.033125487
denmark	0.033125487
information analysis	0.033125487
vehicle to vehicle communications	0.033125487
covariance matrix	0.033125487
eco routing	0.033125487
fog	0.032735776
systematic literature review	0.032735776
predictive control systems	0.032735776
autonomous vehicle	0.032735776
governance approach	0.032346064
infrastructure systems	0.032346064
segmentation	0.032346064
cities	0.032346064
engineering education	0.032346064
fossil fuels	0.032346064
localisation	0.032346064
sensor networks	0.032346064
electric loads	0.032346064
photovoltaic system	0.032346064
social networking (online)	0.031956352
complex networks	0.031956352
facilities management	0.031956352

model and simulation	0.031956352
decision supports	0.031956352
humidity control	0.031956352
cross-domain	0.031566641
digital elevation model	0.031566641
information systems	0.031566641
spain	0.031566641
reconstruction	0.031566641
high resolution	0.031176929
architecture	0.031176929
drones	0.031176929
solar power generation	0.031176929
urban morphology	0.030787217
in-buildings	0.030787217
rapid urbanizations	0.030787217
knowledge	0.030397506
mixed reality	0.030397506
budget control	0.030397506
quality control	0.030397506
renewable energy integrations	0.030397506
solar power	0.030397506
textures	0.030397506
computer software	0.030007794
manufacturing	0.030007794
electric load dispatching	0.030007794
structural optimization	0.030007794
urban mobility	0.030007794
airflow	0.030007794
agriculture	0.029618083
classification (of information)	0.029618083
disaster management	0.029618083
digitalization	0.029618083
infrastructure planning	0.029618083
recurrent neural networks	0.029618083
stormwaters	0.029618083
zero emission	0.029618083
simulation model	0.029618083
videorecording	0.029618083
decarbonisation	0.029228371
real time monitoring	0.029228371
conceptual model	0.029228371
support vector machines	0.029228371
bayesian analysis	0.029228371
hydraulic models	0.029228371
models	0.029228371
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theoretical model	0.029228371
wind power	0.028838659

traffic simulations	0.028838659
urban resilience	0.028838659
catchments	0.028838659
convolution	0.028838659
cost reduction	0.028838659
highway planning	0.028838659
image fusion	0.028838659
water distribution systems	0.028838659
nondestructive examination	0.028838659
scheduling	0.028448948
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planning	0.028448948
renewable resource	0.028448948
microclimate	0.028448948
open systems	0.028448948
thermography	0.028448948
digital mapping	0.028059236
decentralised	0.028059236
federated learning	0.028059236
fault detection	0.028059236
smart solutions	0.028059236
travel time	0.028059236
water networks	0.028059236
humidity	0.028059236
markov chain	0.028059236
monitoring system	0.027669525
project management	0.027669525
quality of service	0.027669525
greenhouse gases	0.027669525
laser applications	0.027669525
mesh generation	0.027669525
tunnel	0.027669525
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airborne lidar	0.027279813
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accuracy assessment	0.027279813
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building stocks	0.026890101
large scale systems	0.026890101
photogrammetry	0.026890101
resilience	0.026890101
motion planning	0.026890101
wireless sensor networks	0.026890101
condition assessments	0.026890101
measurement accuracy	0.026890101
health care	0.026500390

city information modeling	0.026500390
integrated approach	0.026500390
mobile mapping systems	0.026500390
waters managements	0.026500390
probability density function	0.026500390
traffic and transport	0.026500390
low-costs	0.026110678
neighbourhood	0.026110678
simple++	0.026110678
south korea	0.026110678
inspection	0.026110678
greenspace	0.025720966
clustering algorithms	0.025720966
electric vehicles	0.025720966
investments	0.025720966
urban building energy modeling	0.025720966
wireless networks	0.025720966
nature-based solution	0.025720966
air conditioning	0.025331255
dynamics	0.025331255
fog computing	0.025331255
geo-spatial data	0.025331255
potential energy	0.025331255
sustainable energy	0.025331255
water budget	0.025331255
maintenance	0.024941543
infrastructural development	0.024941543
urban infrastructure	0.024941543
digital twin technology	0.024941543
electric power transmission networks	0.024941543
inference engines	0.024941543
objects detection	0.024941543
public-private partnership	0.024941543
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virtual environments	0.024551832
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waste management	0.024162120
6g mobile communication	0.024162120
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energy efficient	0.024162120
cyber-attacks	0.024162120

error analysis	0.024162120
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high-fidelity	0.023772408
industrial robots	0.023772408
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3d point cloud	0.023382697
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integer programming	0.022992985
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state of the art	0.022603274
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intrusion detection systems	0.022603274
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air mobility	0.022213562
level-of-detail	0.022213562
medical services	0.022213562
object recognition	0.022213562
unmanned vehicle	0.022213562

hong kong	0.022213562
participatory approach	0.021823850
bibliometric	0.021823850
building facades	0.021823850
carbon footprint	0.021823850
digital representations	0.021823850
ecosystem	0.021823850
sustainable city	0.021823850
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computational fluid dynamics	0.021434139
peer to peer networks	0.021434139
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vibration	0.021434139
water conservation	0.021434139
water pollution	0.021434139
air pollutant	0.021434139
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mobile	0.003897116
model learning	0.003897116
privacy enhancing technologies	0.003897116
privacy preserving	0.003897116
privacy-preserving techniques	0.003897116
psychology	0.003897116
rainwater	0.003897116
refugee	0.003897116
seoul [seoul (ads)]	0.003897116
seoul [south korea]	0.003897116
terrain	0.003897116
action plan	0.003507405
biofuel	0.003507405
citizen engagements	0.003507405
data usage	0.003507405
delta	0.003507405
extraction method	0.003507405
hamburg [hamburg (stt)]	0.003507405
health worker	0.003507405
new zealand	0.003507405
nigeria	0.003507405
optical and wireless	0.003507405
pearl river [united states]	0.003507405
regional development	0.003507405
regional economy	0.003507405
rotterdam	0.003507405
rural development	0.003507405

```
rural economy                                                    0.003507405
rural planning                                                   0.003507405
signal processing                                                0.003507405
south holland                                                    0.003507405
spatial variation                                                0.003507405
spectral analysis                                                0.003507405
state of research                                                0.003507405
temporal variation                                               0.003507405
urban-rural migration                                            0.003507405
zhujiang delta                                                   0.003507405
analytical framework                                             0.003117693
and internet of thing                                            0.003117693
cartography                                                      0.003117693
freight transport                                                0.003117693
gartner                                                          0.003117693
grassroots level                                                 0.003117693
land management                                                  0.003117693
land market                                                      0.003117693
land tenure                                                      0.003117693
nonlinearity                                                     0.003117693
pakistan                                                         0.003117693
scientific knowledge                                             0.003117693
systems implementation                                           0.003117693
technological trends                                             0.003117693
alternative agriculture                                          0.002727981
curriculum                                                       0.002727981
education                                                        0.002727981
exploration                                                      0.002727981
fine grained sediment                                            0.002727981
gas industry                                                     0.002727981
laser method                                                     0.002727981
singapore [southeast asia]                                       0.002727981
student                                                          0.002727981
teaching                                                         0.002727981
automobile industry                                              0.002338270
classification                                                   0.002338270
computer networks                                                0.002338270
diffusion                                                        0.002338270
imaging method                                                   0.002338270
language                                                         0.002338270
methodology                                                      0.002338270
phase transition                                                 0.002338270
regeneration                                                     0.002338270
semantic standardization                                         0.002338270

$params
                params                                    values
1                    n                                        40
```

```
2                Title Co-ocorrência de Palavras-chave do Autor
3                 type                                fruchterman
4                label                                       TRUE
5            labelsize                                          1
6            label.cex                                       TRUE
7          label.color                                      FALSE
8                 halo                                      FALSE
9              cluster                                   walktrap
10 community.repulsion                                        0.1
11                size                                          3
12            size.cex                                       TRUE
13              curved                                      FALSE
14             noloops                                       TRUE
15     remove.multiple                                      FALSE
16     remove.isolates                                      FALSE
17            edgesize                                          3
18           edges.min                                          3
19               alpha                                        0.5
20             verbose                                       TRUE
```

```r
dev.off()
```

```
pdf
  2
```

```r
cat("Rede salva em: output/figures/09_coocorrencia_keywords.png\n")
```

```
Rede salva em: output/figures/09_coocorrencia_keywords.png
```

5 Mapa Temático (Thematic Map)

O **mapa temático** é a análise mais importante deste artigo. Ele posiciona os temas do campo em quatro quadrantes baseados em dois eixos:

- **Centralidade (eixo X):** Quão importante o tema é para o campo como um todo. Alta centralidade = tema central e bem conectado com outros temas.
- **Densidade (eixo Y):** Quão desenvolvido/maduro o tema está internamente. Alta densidade = tema coeso, com forte estrutura interna.

5.1 Os 4 Quadrantes do Mapa Temático

```
                    Alta Densidade
                         |
    TEMAS NICHO          |      TEMAS MOTORES
    (bem desenvolvidos   |      (centrais e bem
    mas periféricos)     |      desenvolvidos)
                         |

                         |
    TEMAS EMERGENTES     |      TEMAS BÁSICOS
```

```
    OU EM DECLÍNIO       |      (centrais mas ainda
    (periféricos e       |      pouco desenvolvidos)
    pouco desenvolvidos) |
                         |
                    Baixa Densidade
    Baixa Centralidade      Alta Centralidade
```

Interpretação para o artigo:

- Se "decision support" / "governance" estiver no quadrante **emergente/declínio**: O tema existe mas é periférico e pouco desenvolvido → confirma o gap
- Se estiver **ausente** do mapa: O tema nem sequer é reconhecido como cluster relevante → gap ainda mais forte
- Se estiver como **tema básico**: É reconhecido como importante mas pouco desenvolvido → oportunidade de pesquisa clara

```r
# Gerar o mapa temático usando keywords do autor
# field = "DE" usa as Author Keywords
# n = número mínimo de ocorrências para incluir a keyword
# minfreq = frequência mínima para incluir no mapa
mapa <- thematicMap(
  df,
  field = "DE",          # Author Keywords
  n = 250,               # Número de termos a considerar
  minfreq = 3,           # Frequência mínima para inclusão
  stemming = FALSE,      # Não aplicar stemming (manter termos originais)
  size = 0.5,            # Tamanho dos rótulos
  repel = TRUE           # Evitar sobreposição de rótulos
)

# O objeto retornado contém o plot e os dados dos clusters
print(mapa$map)
```

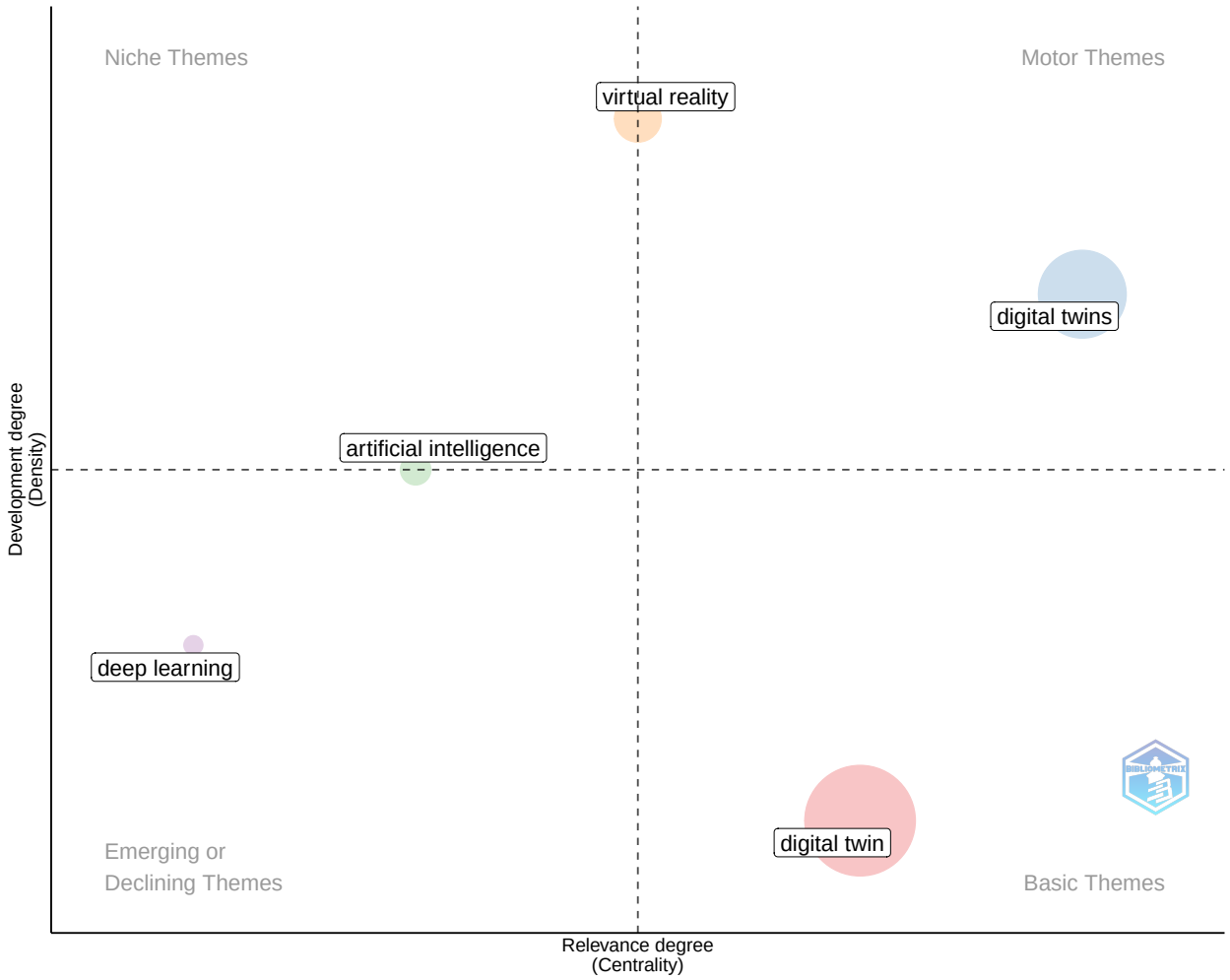


Figure 3: Mapa temático — posicionamento dos temas do campo em quatro quadrantes

```
# Salvar o mapa temático em alta resolução
ggsave(
  here("output", "figures", "10_mapa_tematico.png"),
  plot = mapa$map,
  width = 12, height = 10, dpi = 300
)

cat("Mapa temático salvo em: output/figures/10_mapa_tematico.png\n")
```

```
Mapa temático salvo em: output/figures/10_mapa_tematico.png
```

5.2 Detalhamento dos Clusters do Mapa Temático

Vamos examinar a composição de cada cluster para identificar onde os temas de interesse (decision support, governance) se posicionam.

```
# Extrair informações dos clusters
clusters_info <- mapa$clusters
```

```r
cat("=== CLUSTERS DO MAPA TEMÁTICO ===\n\n")
```

```
=== CLUSTERS DO MAPA TEMÁTICO ===
```

```r
# Iterar sobre cada cluster e mostrar seus termos
for (i in seq_len(nrow(clusters_info))) {
  # Determinar o quadrante baseado na centralidade e densidade
  centralidade <- clusters_info$rcentrality[i]
  densidade <- clusters_info$rdensity[i]

  quadrante <- case_when(
    centralidade >= 0.5 & densidade >= 0.5 ~ "MOTOR",
    centralidade < 0.5 & densidade >= 0.5 ~ "NICHO",
    centralidade >= 0.5 & densidade < 0.5 ~ "BÁSICO",
    TRUE ~ "EMERGENTE/DECLÍNIO"
  )

  cat(sprintf("Cluster '%s' [%s]\n", clusters_info$name[i], quadrante))
  cat(sprintf("  Centralidade: %.3f | Densidade: %.3f\n",
              centralidade, densidade))
  cat(sprintf("  Palavras: %s\n\n", clusters_info$words[i]))
}
```

```
Cluster 'digital twin' [MOTOR]
  Centralidade: 4.000 | Densidade: 1.000
  Palavras: digital twin 233
smart city 72
urban planning 23
machine learning 22
simulation 17
big data 12
3d city model 10
building information modeling (bim) 7
built environment 7
industry 4.0 7

Cluster 'digital twins' [MOTOR]
  Centralidade: 5.000 | Densidade: 4.000
  Palavras: digital twins 109
smart cities 57
internet of things 26
sustainability 16
iot 15
bim 13
building information modeling 7
lidar 7
ai 6
gis 6
```

```
Cluster 'artificial intelligence' [MOTOR]
  Centralidade: 2.000 | Densidade: 3.000
  Palavras: artificial intelligence 31
urban digital twin 14
geographic information system 7
remote sensing 5
standards 5
traffic simulation 5
edge computing 4
living lab 4
platform 4
sustainable urban development 4

Cluster 'deep learning' [MOTOR]
  Centralidade: 1.000 | Densidade: 2.000
  Palavras: deep learning 27
semantic segmentation 9
city digital twin 7
digitalization 7
urban digital twins 7
point cloud 5
3d reconstruction 4
point clouds 4
3d city models 3
attention mechanism 3

Cluster 'virtual reality' [MOTOR]
  Centralidade: 3.000 | Densidade: 5.000
  Palavras: virtual reality 18
blockchain 16
digital twin (dt) 15
metaverse 15
internet of things (iot) 12
security 8
5g 7
climate change 7
artificial intelligence (ai) 6
augmented reality (ar) 5
```

```r
# Exportar dados dos clusters para análise detalhada
write_csv(clusters_info, here("output", "tables", "clusters_mapa_tematico.csv"))
cat("Dados dos clusters exportados para: output/tables/clusters_mapa_tematico.csv\n")
```

```
Dados dos clusters exportados para: output/tables/clusters_mapa_tematico.csv
```

6 Evolução Temática (Thematic Evolution)

Análise de como os temas evoluíram ao longo do tempo, dividindo o período em subperíodos para observar mudanças.

```
# Dividir o período em fatias temporais para análise de evolução
# Sugestão: 2015-2019 (fase inicial) e 2020-2024 (fase de crescimento)
evolucao <- thematicEvolution(
  df,
  field = "DE",
  years = c(2019),       # Ponto de corte entre os períodos
  n = 100,               # Número de termos por período
  minFreq = 3            # Frequência mínima
)

# Plotar o Sankey diagram da evolução temática
plotThematicEvolution(evolucao$Nodes, evolucao$Edges)
```

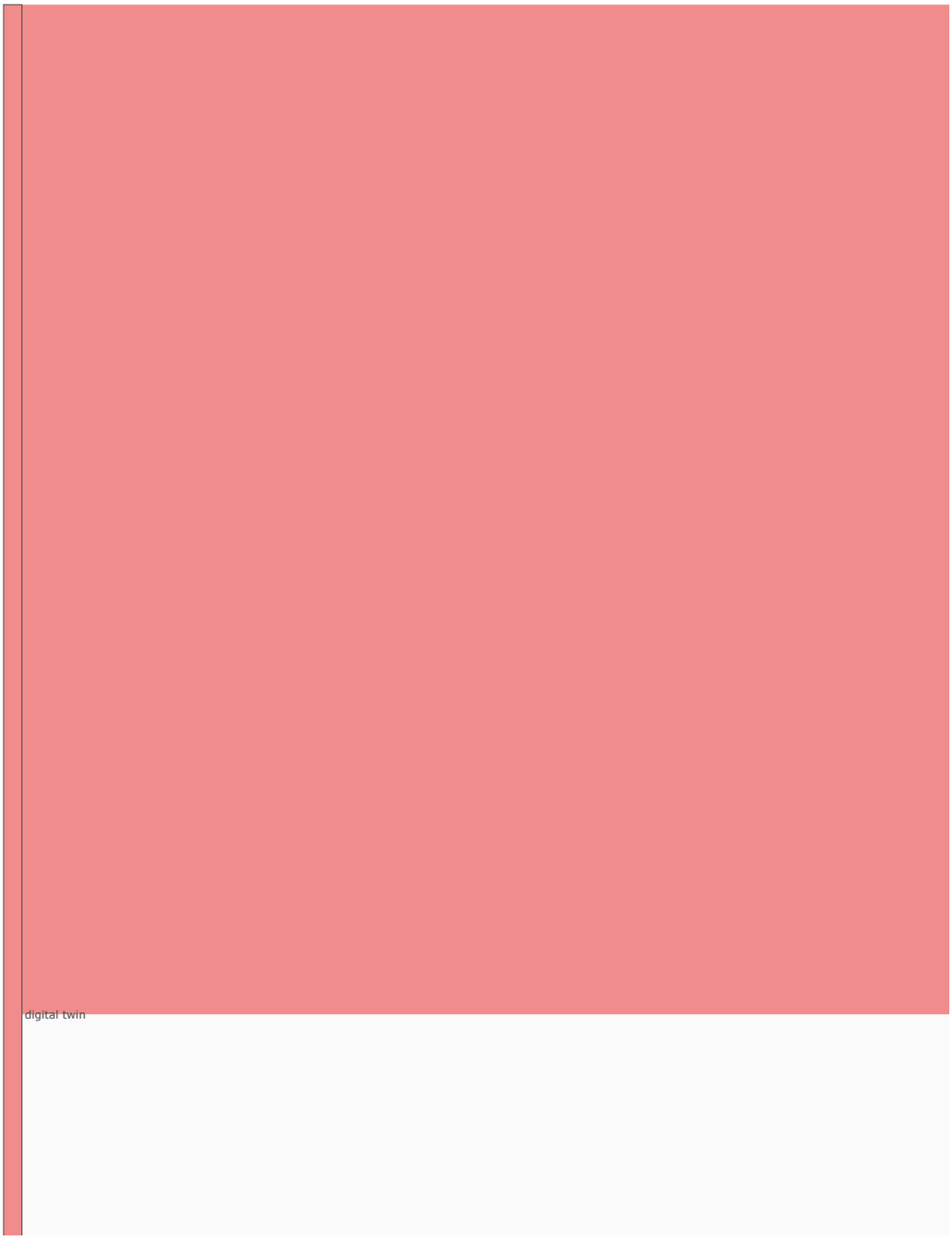
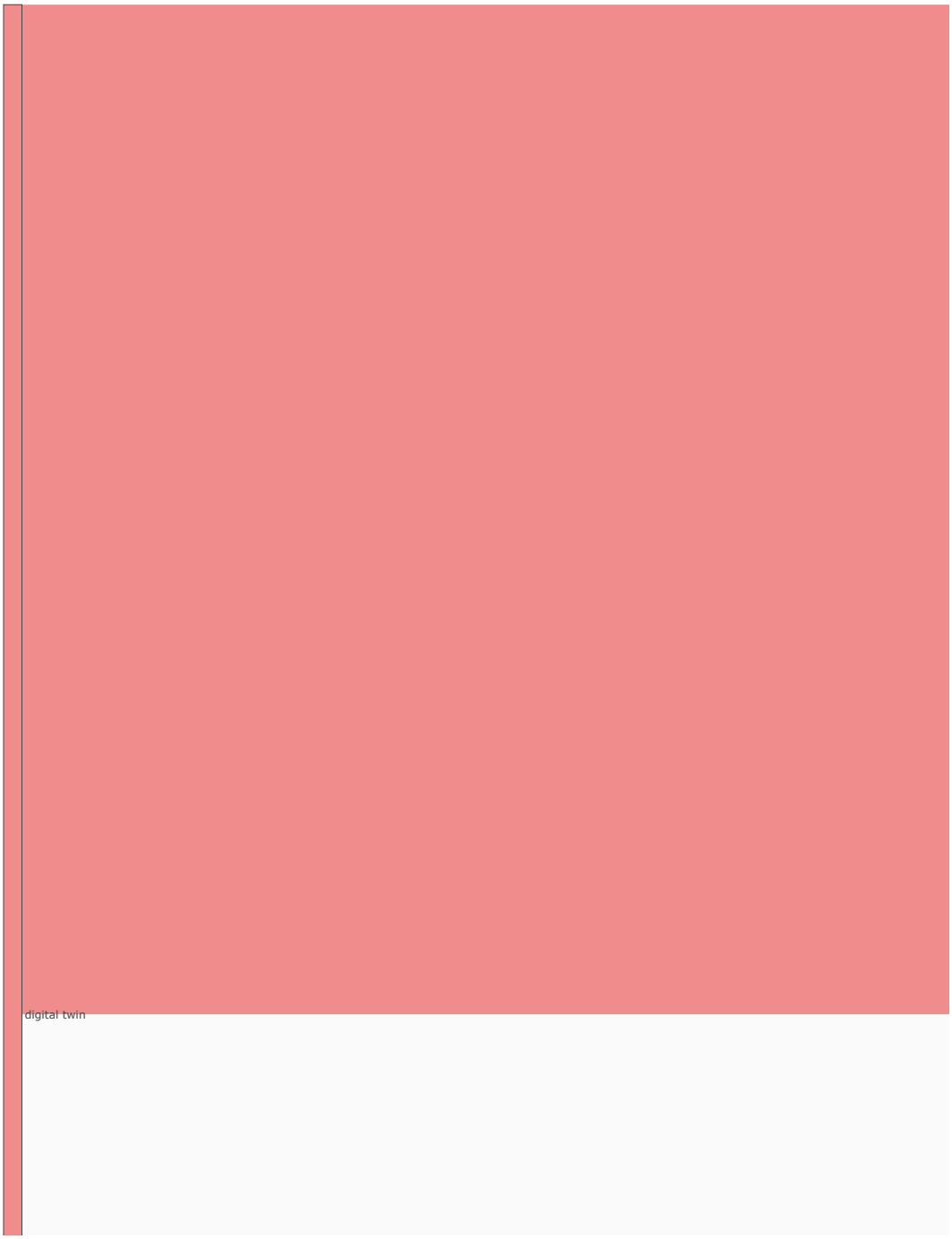


Figure 4: Evolução temática ao longo do período de estudo

```r
# Salvar a evolução temática
png(here("output", "figures", "11_evolucao_tematica.png"),
    width = 14, height = 8, units = "in", res = 300)

plotThematicEvolution(evolucao$Nodes, evolucao$Edges)
```




```
dev.off()
```

```
pdf
  2
```

```r
cat("Evolução temática salva em: output/figures/11_evolucao_tematica.png\n")
```

```
Evolução temática salva em: output/figures/11_evolucao_tematica.png
```

7 Resumo da Análise de Keywords

As análises deste notebook revelam:

1. **Keywords dominantes:** Os termos mais frequentes indicam os focos principais da pesquisa (provavelmente IoT, BIM, simulation, etc.)
2. **Clusters temáticos:** A rede de co-ocorrência mostra como os temas se agrupam, revelando subcampos distintos
3. **Mapa temático:** Posiciona cada tema em relação à sua centralidade e maturidade no campo. **Esta é a evidência principal do artigo.**
4. **Gap identificado:** A posição (ou ausência) de termos como "decision support" e "public management" no mapa temático sustenta a principal contribuição do artigo.

   Para o artigo: O mapa temático deve ser discutido extensivamente na seção de Discussion, conectando a posição dos temas ao argumento de que a dimensão de suporte à decisão gerencial pública é um gap significativo na literatura.

   Próximo passo: O notebook `05_results_summary.qmd` compila todos os resultados em um formato adequado para o manuscrito.